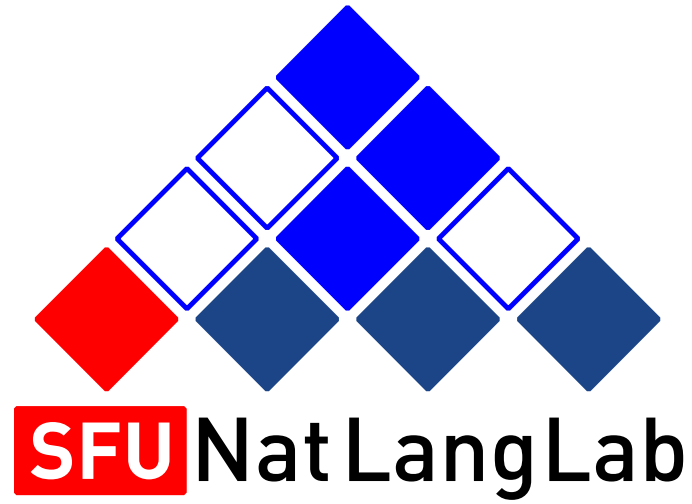




CMPT 413/713: Natural Language Processing

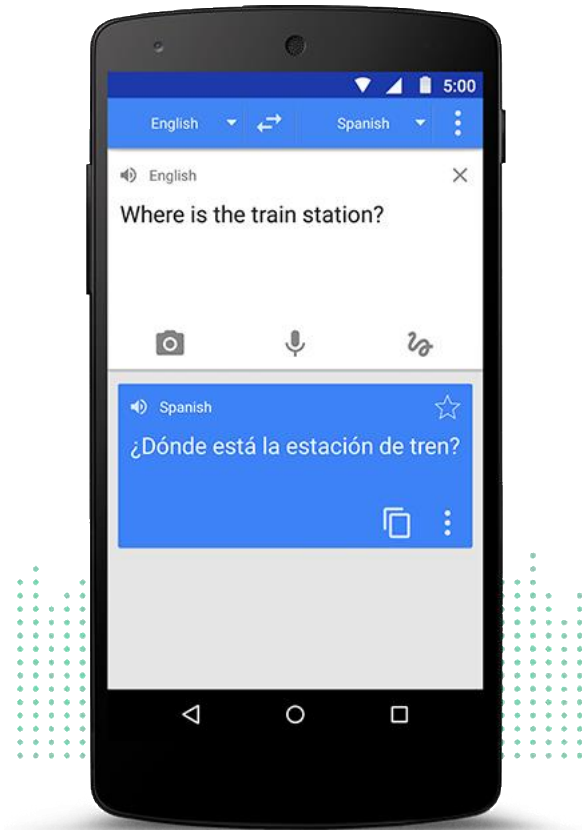
Angel Xuan Chang

2026-01-05

Today

- Intro to NLP
 - What is language? NLP vs computational linguistics.
 - Brief history of NLP
 - Levels of representation in language
 - Key questions of NLP
- Course Logistics
- Intro to Language Modeling

NLP is everywhere



Google translate



Virtual assistants

NLP is everywhere!



ChatGPT can answer all kinds of questions!

Can you explain how LLMs work?

Sure! Here's a clear, intuitive explanation of how Large Language Models (LLMs) work:

What an LLM is

A Large Language Model is a type of artificial neural network trained to understand and generate human-like text. Models like GPT learn patterns in language from massive amounts of text data.

How an LLM works (step by step)

1. Training on huge datasets

LLMs read billions of words—from books, websites, articles, etc.—and learn:

- grammar
- facts about the world
- writing styles
- how ideas relate to each other

They don't store exact copies of the text; instead, they learn statistical patterns and how words work together.

2. Transformers: the core architecture

Modern LLMs use a structure called a **Transformer**, which has two key ideas:

Attention

Attention lets the model look at all words in a sentence at once and decide which words are most important to predict the next word.

Example:

In the sentence "The cat that the dog chased escaped," attention helps the model connect "cat" ↔ "escaped" even though they are far apart.

Layers

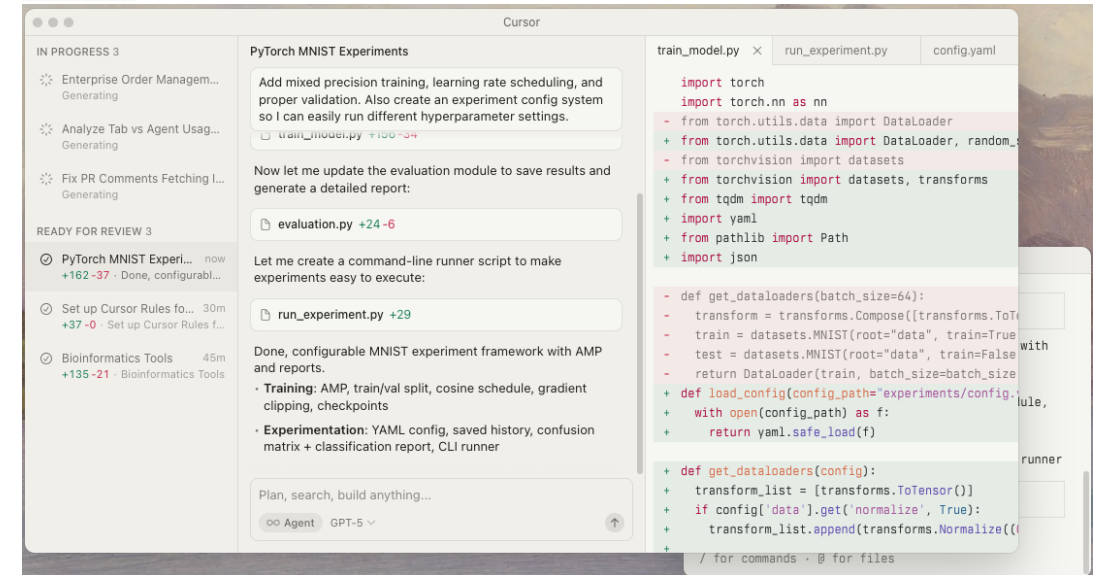
Transformers stack many attention layers.

Each layer learns more abstract patterns—first grammar, then meaning, then reasoning structures.

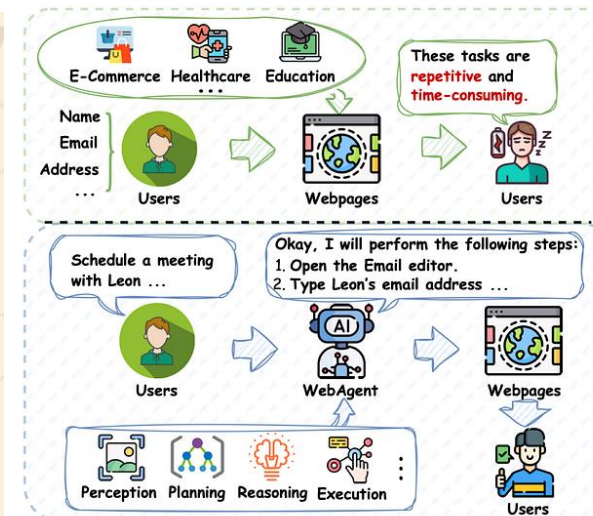
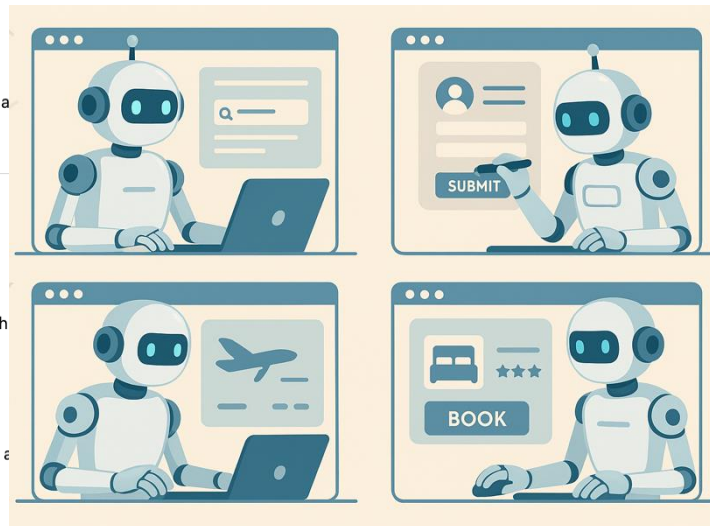


Cursor for coding

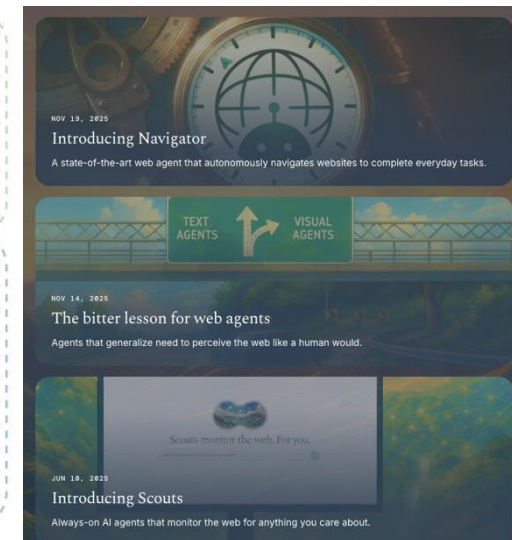
<https://cursor.com/>



LLM agents to automate tasks



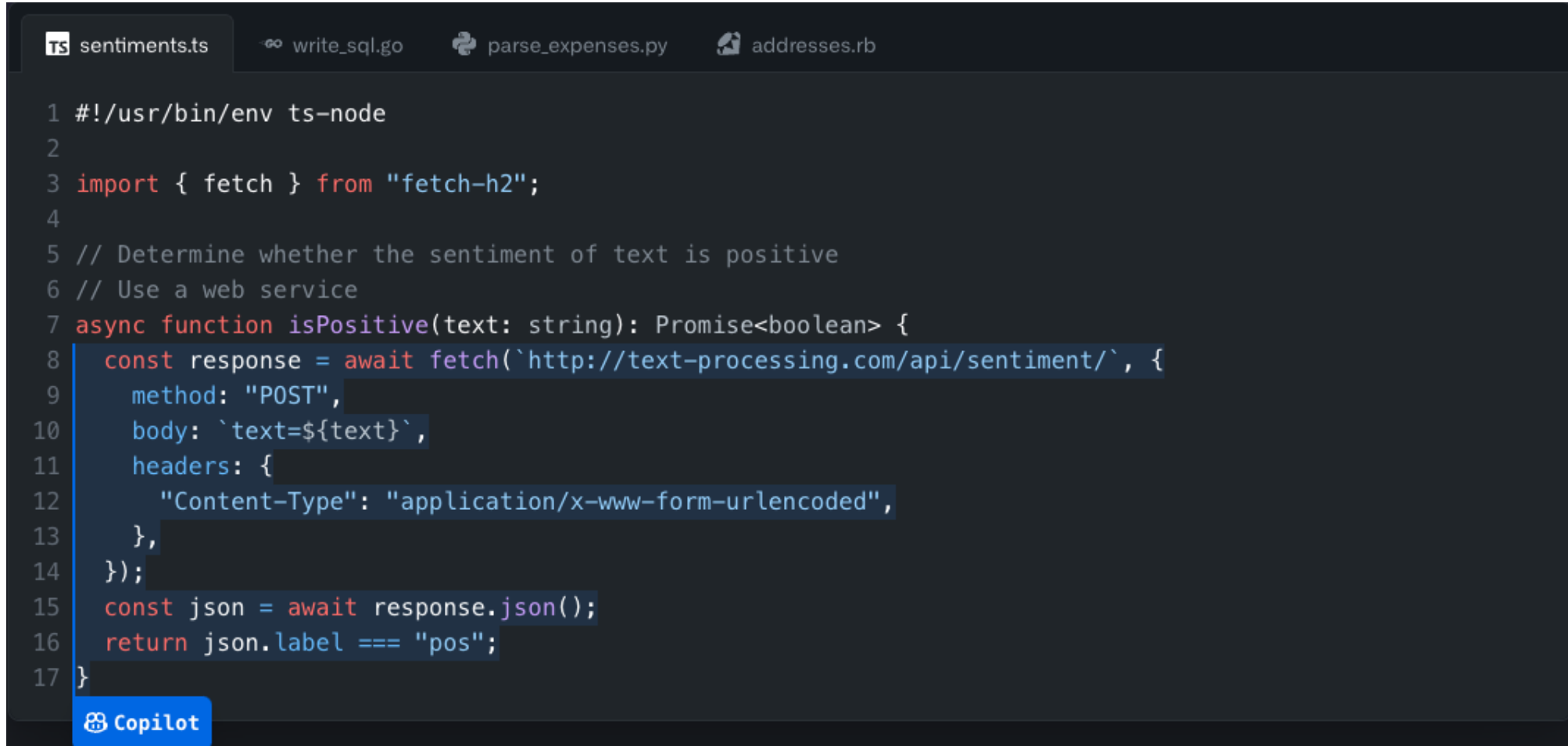
<https://arxiv.org/pdf/2503.23350>



Yutori

Copilot

Generate code from text!

A screenshot of a code editor interface with a dark theme. At the top, there are four tabs: 'sentiments.ts' (active), 'write_sql.go', 'parse_expenses.py', and 'addresses.rb'. The main editor area displays TypeScript code for a function named 'isPositive'. The code is line-numbered from 1 to 17. A blue selection bar highlights the code from line 8 to line 16. At the bottom left of the editor, there is a blue button with the Copilot logo and the text 'Copilot'.

```
1 #!/usr/bin/env ts-node
2
3 import { fetch } from "fetch-h2";
4
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
```

<https://github.com/features/copilot>

Text to image generation

 **OpenAI** DALL-E 2

a painting of a fox sitting in a field at sunrise in the style of Claude Monet



<https://openai.com/dall-e-2/>

 **Imagen**

A cute sloth holding a small treasure chest.
A bright golden glow is coming from the chest.



<https://imagen.research.google/>

Try it out yourself: <https://huggingface.co/spaces/stabilityai/stable-diffusion>,
<https://www.craiyon.com/>, <https://www.midjourney.com/home/>

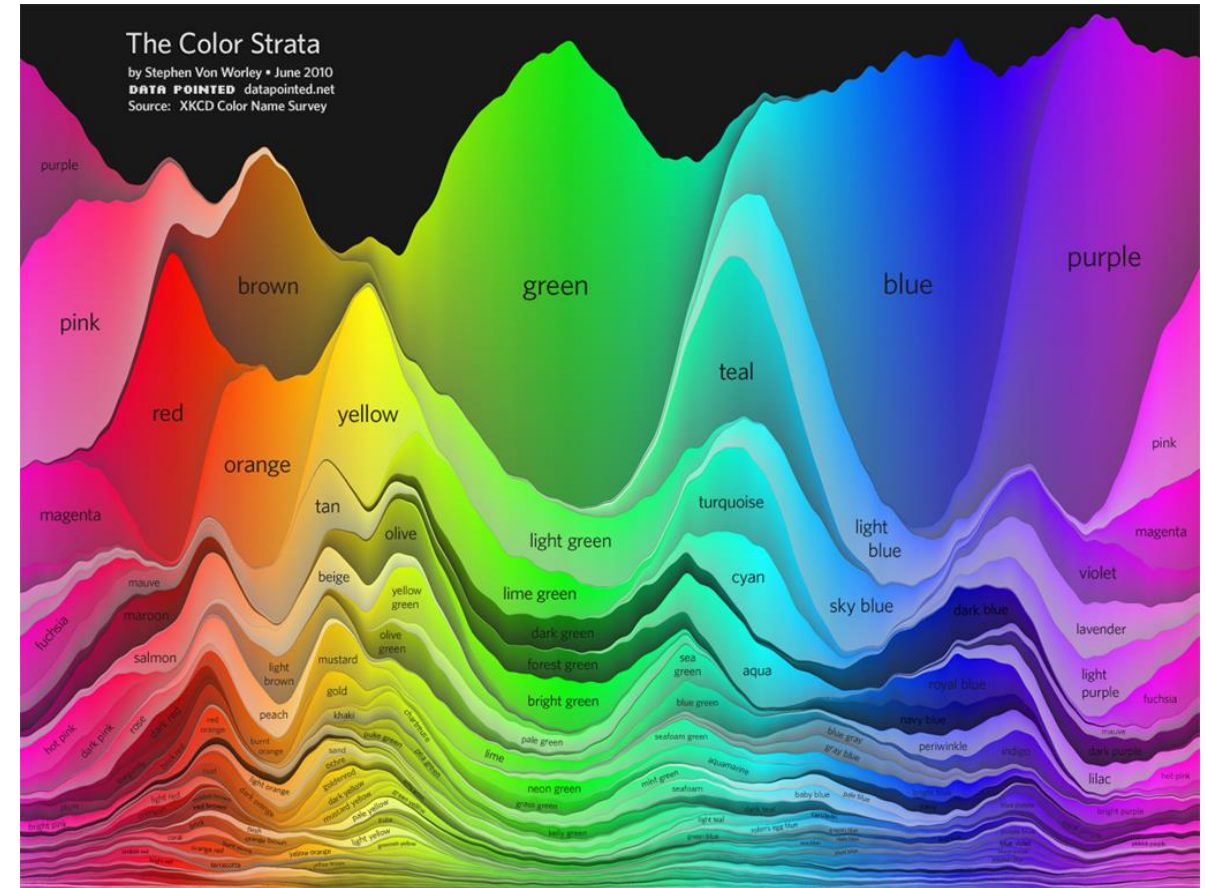
What is language?

- Language is used to communicate
 - Things, actions, abstract concepts



What is language?

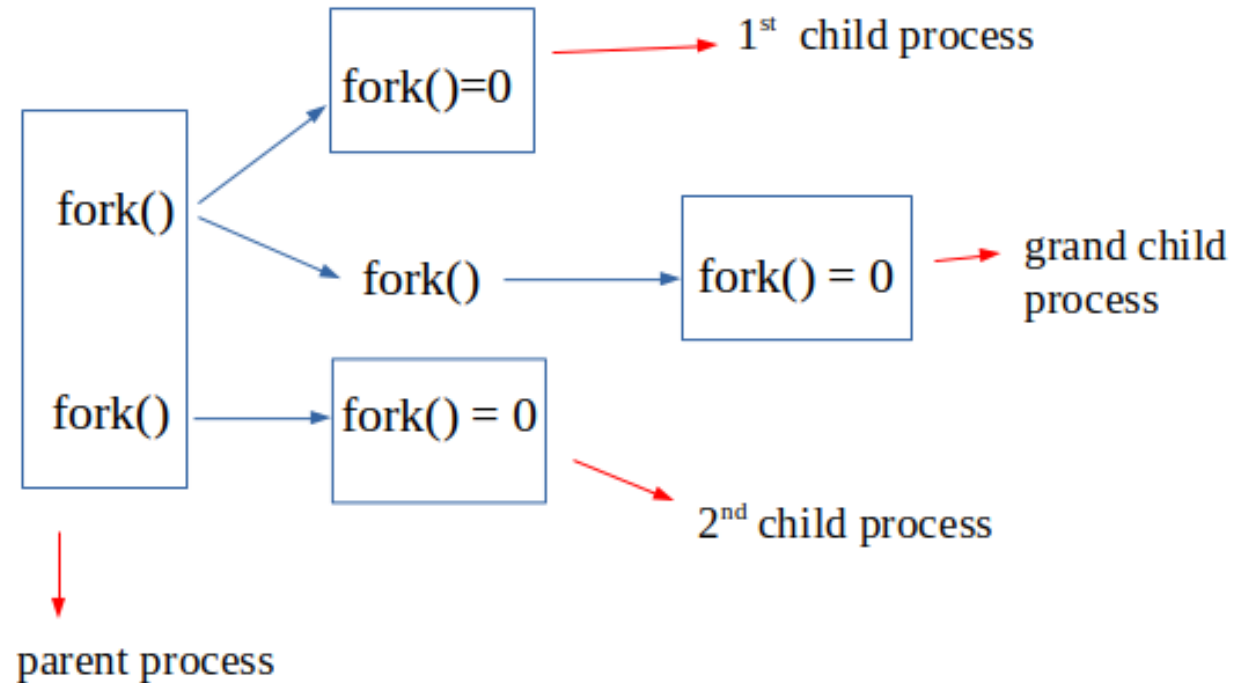
- Language puts categories on the world
 - It discretizes a continuous space



What is language?

- Language picks out what is salient and important
- What concepts do we have words for?

fork



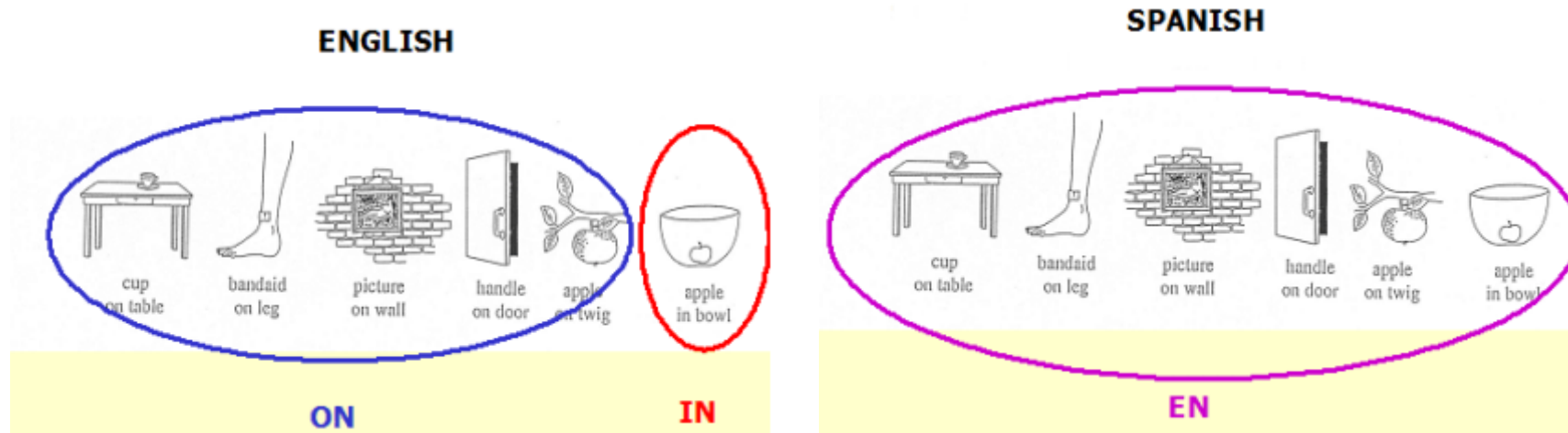
What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



<http://pyersqr.org/classes/Ling731/Space2.htm>

Natural Language Processing



Building **useful system** to process language

Computational Linguistics

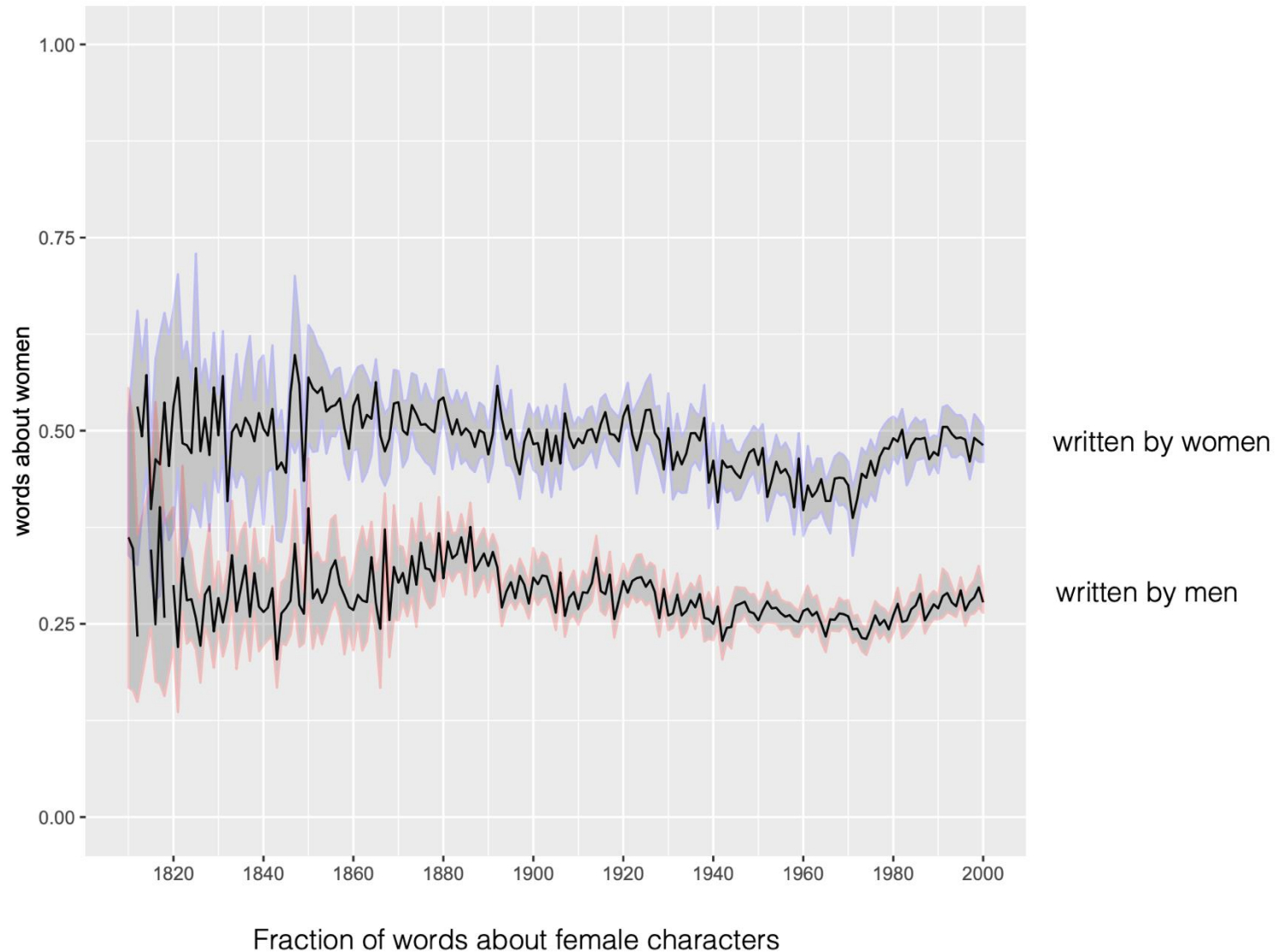


"I just had a heated conversation
with my computer."

(image credit: <https://www.enterrasolutions.com/blog/computational-linguistics-and-natural-language-processing/>)

Using computers to study **human** language

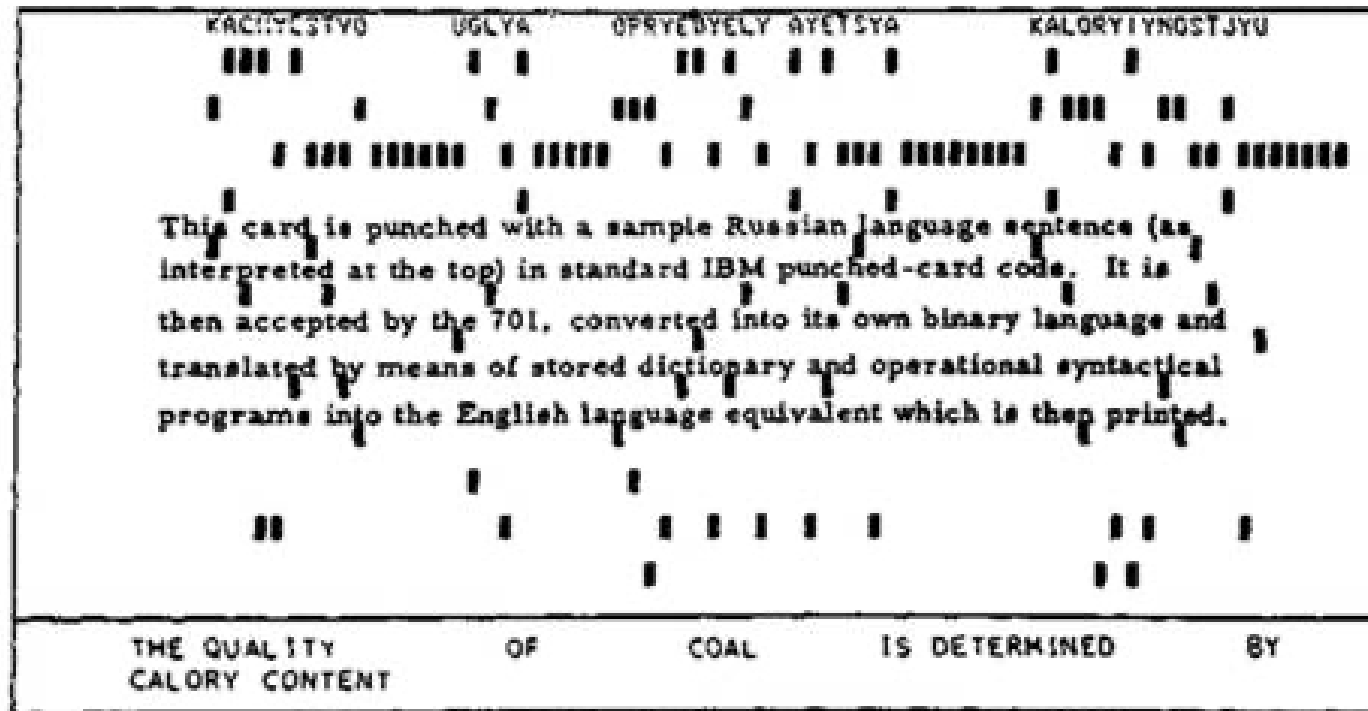
Analyzing word usage in literature



Ted Underwood, David Bamman, and Sabrina Lee (2018),
"The Transformation of Gender in English-Language Fiction," Cultural Analytics

Brief history of NLP

Beginnings



Specimen punched card and below a strip with translation, printed within a few seconds

Georgetown-
IBM
experiment,
1954

“Within three or five years, machine translation will be a solved problem”

SHRDLU (Winograd, 1968)

Video of actual system: <https://www.youtube.com/watch?v=bo4RvYJYOzI>

Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Person (changing their mind): Find a block which is taller than the one you are holding and put it into the box.

Computer: By "it", I assume you mean the block which is taller than the one I am holding.

Computer: OK.

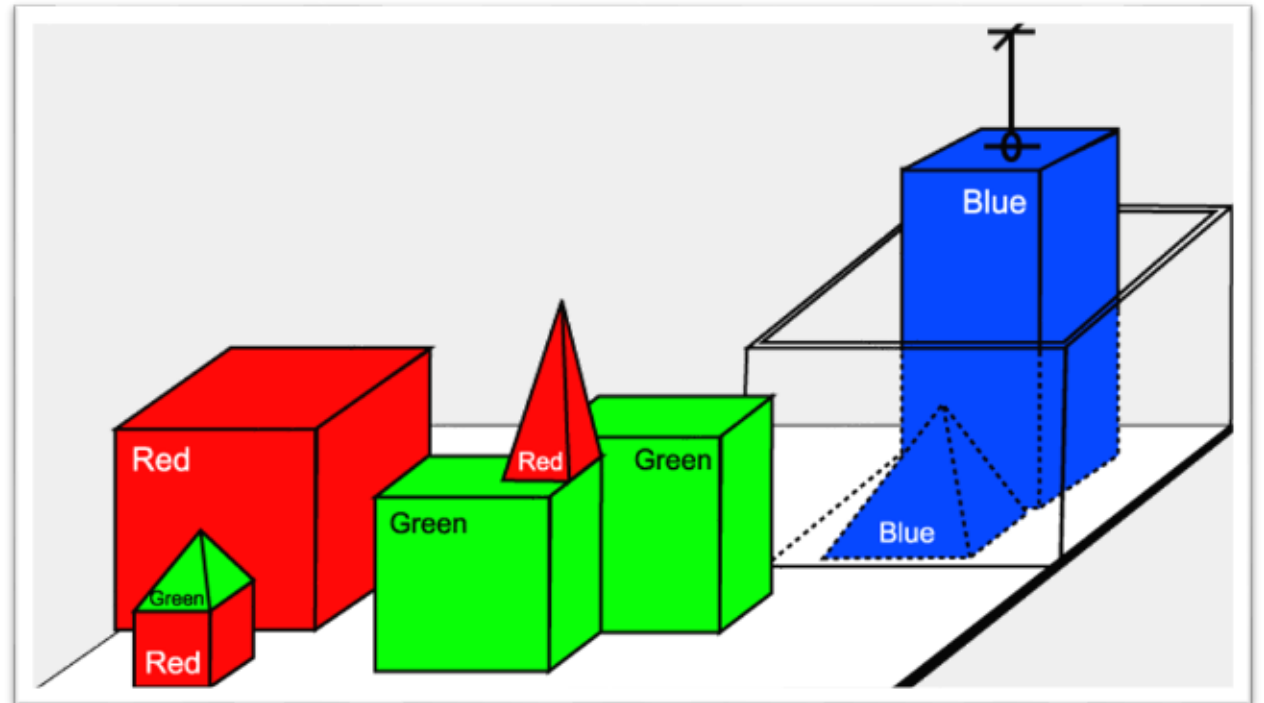
Person: What does the box contain?

Computer: The blue pyramid and the blue block.

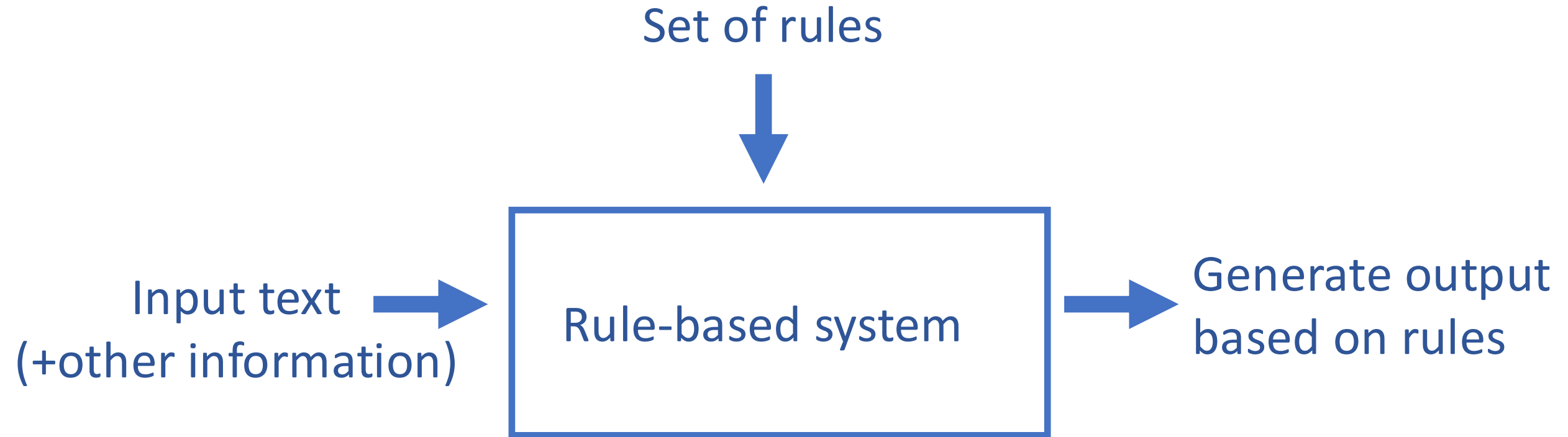
Person: What is the pyramid supported by?

Computer: The box.

Lots of rules!



Using rules for NLP



Match a pattern

- Regular expressions

 - . *ed => past tense verb

- Contain keyword: delicious

Eliza (Weizenbaum, 1966)

Demo: <http://psych.fullerton.edu/mbirnbaum/psych101/Eliza.htm>

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

Rogarian psychologist:
reflect back what the patient said

- Set of ranked transformation **rules** based on **keywords**
- Apply rules based on matching words in sentence

Transformation rules

Keywords are ranked from specific to general

I know everybody laughed at me

- “I” is a very general keyword:

I: (I *) -> (You say you 2)

YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

- “Everybody” is much more interesting (someone using universals like everybody/always is probably “referring to some quite specific event or person”)

WHO IN PARTICULAR ARE YOU THINKING OF?

- Implementation: keywords stored with their rank

Everybody 5 (*transformation rules*)

I 0 (*transformation rules*)

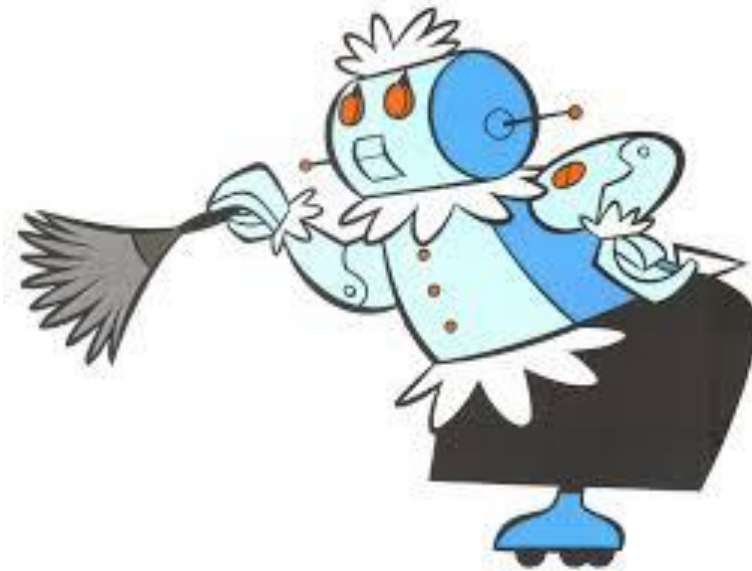
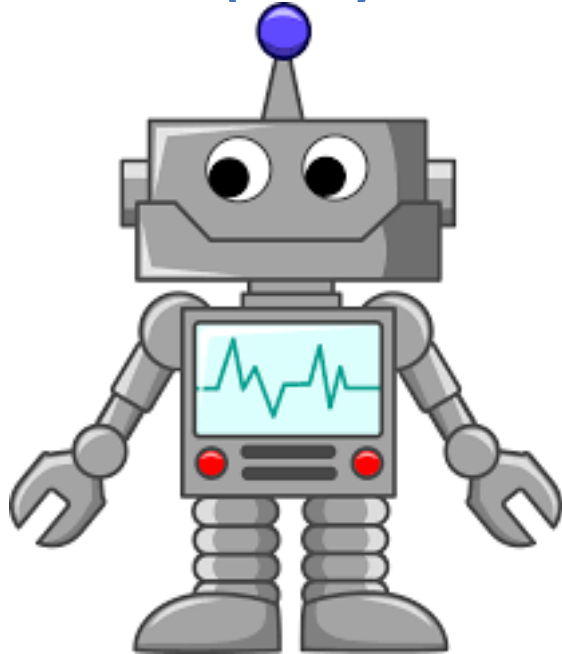
Backoff

Please go on

That's very interesting

I see

Where is my block stacking or
housekeeper robot that I can talk to?
(maybe we are getting close...)



Rosie from the Jetsons



Understanding
language is hard!

Why is natural language processing hard?

Programming Languages

C, C++, Java, Python, ...

- Unambiguous
- Fixed
- Designed
- Learnable?
- Known simple semantics

Natural Languages

French, English, Korean, Chinese, Tagalog, ...

- Ambiguous
- Evolving
- Transmitted
- Learnable
- Complex semantics

Why is NLP hard?

Interpretation of language assumes a common basis of **world knowledge** and **context**

- **Ambiguous:**
 - “bank”, “bat”
 - “Milk Drinkers Turn to Powder”
- **Variation:** Many ways to say same thing
- **Context dependent:**
 - natural language is under-specified



Herb Clark

“bat’k”

Table



Counter

Context-dependence

“I put the bowl on the **table**”



“The numbers in the **table** don’t add up”

Table 8.2 Calculation of the χ^2 test on figures in table 8.1

Class (1)	Expected numbers		O - E		(O-E) ² /E	
	A (2)	B (3)	A (4)	B (5)	A (6)	B (7)
I	11.80	10.20	5.20	-5.20	2.292	2.651
II	24.67	21.33	0.33	-0.33	0.004	0.005
III	39.15	33.85	-0.15	0.15	0.001	0.001
IV	48.81	42.19	-6.81	6.81	0.950	1.009
V	30.57	26.43	1.43	-1.43	0.067	0.077
Total	30.57	134.00	0	0	3.314	3.833

$$\chi^2 = 3.314 + 3.833 = 7.147, df = 4, 0.10 < P < 0.50;$$

“I made her duck”

I cooked her duck (the waterfowl) for dinner

I cooked the duck (the waterfowl) belonging to her.

I created a model of a duck that she now has.

I caused her to lower her head

Some language humor (from real newspaper headlines)

Kids make nutritious snacks

Stolen painting found by tree

Miners refuse to work after death

Squad helps dog bite victim

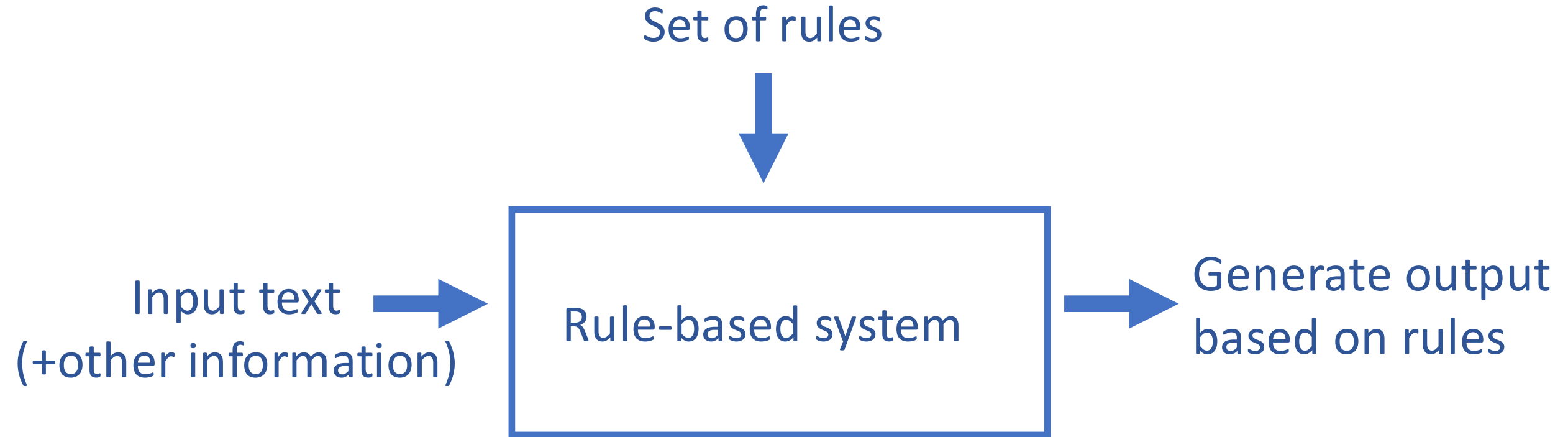
Killer sentenced to die for second time in 10 years

Lack of brains hinders research

Pooches and people digging newly opened park

Using rules for NLP

- Translation with rules started in the 1950s
- Eliza chatbot (1966)
- SHRDLU grounded language understanding (1968)



Rule-based Machine Translation

- Georgetown IBM translator
- Bilingual dictionary to map Russian words to their English counterparts
- But what if the word order is different? How to handle all the nuances of meaning?



1 minute video showing 1954 MT:

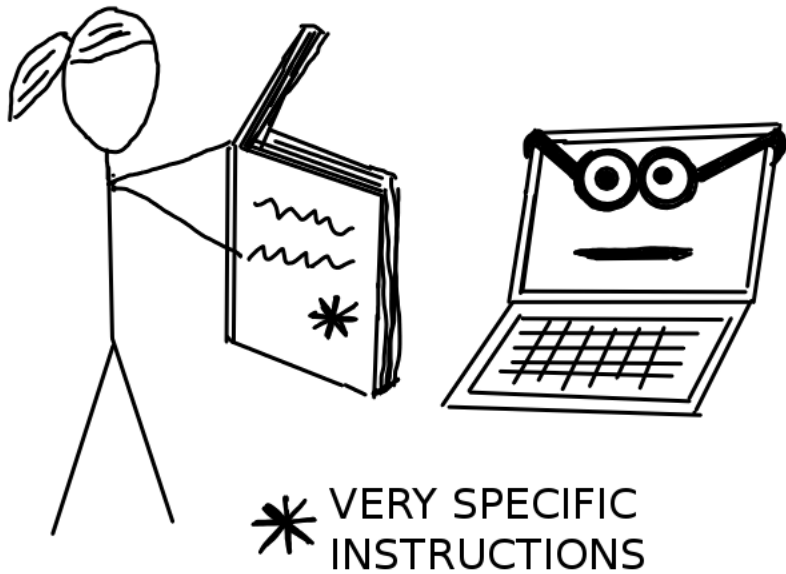
<https://youtu.be/K-HfpsHPmvw>

https://www.ibm.com/ibm/history/exhibits/701/701_translator.html

Coming up rules is hard!

Let's learn from data!

Without Machine Learning



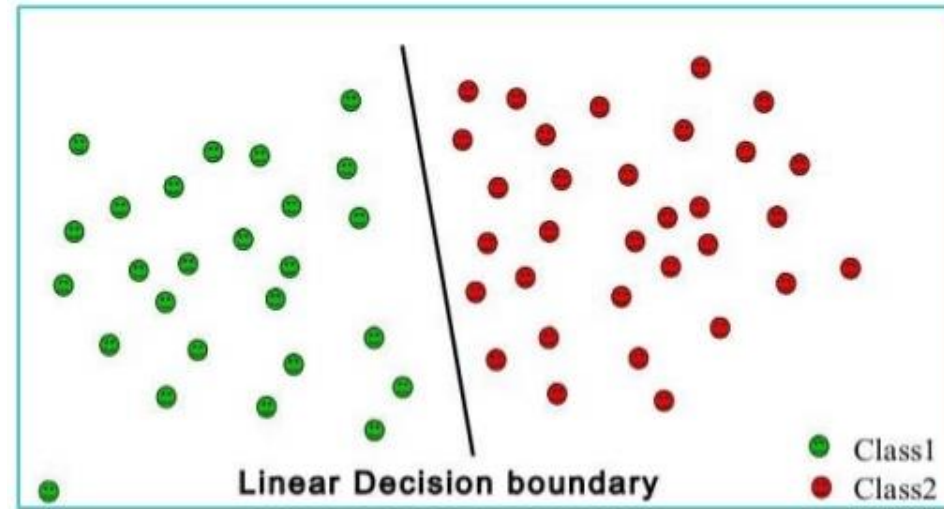
With Machine Learning



<https://christophm.github.io/interpretable-ml-book/terminology.html>

Rise of statistical learning

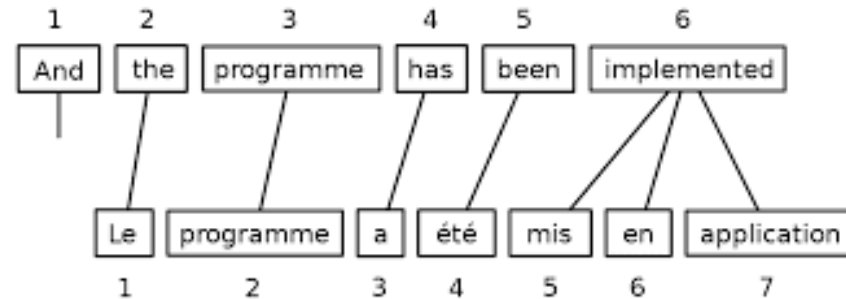
(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT



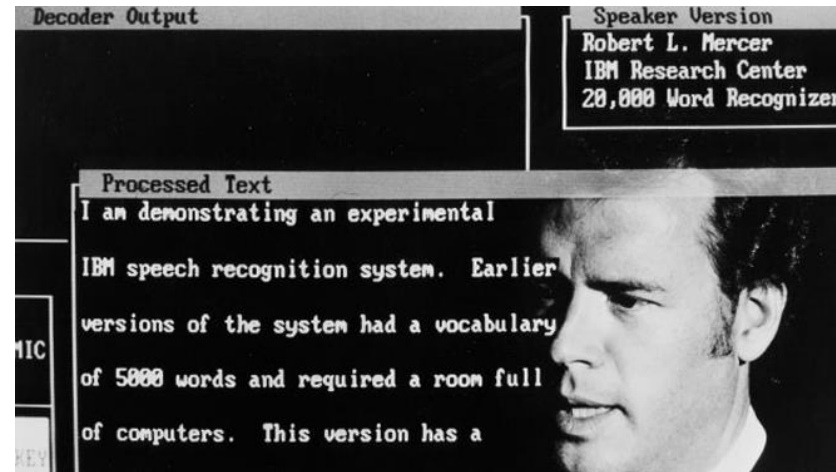
- Use of machine learning techniques in NLP
- Increase in computational capabilities
- Availability of electronic corpora
- Data-driven evaluation of models

Rise of statistical learning

(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT



IBM Models
for translation



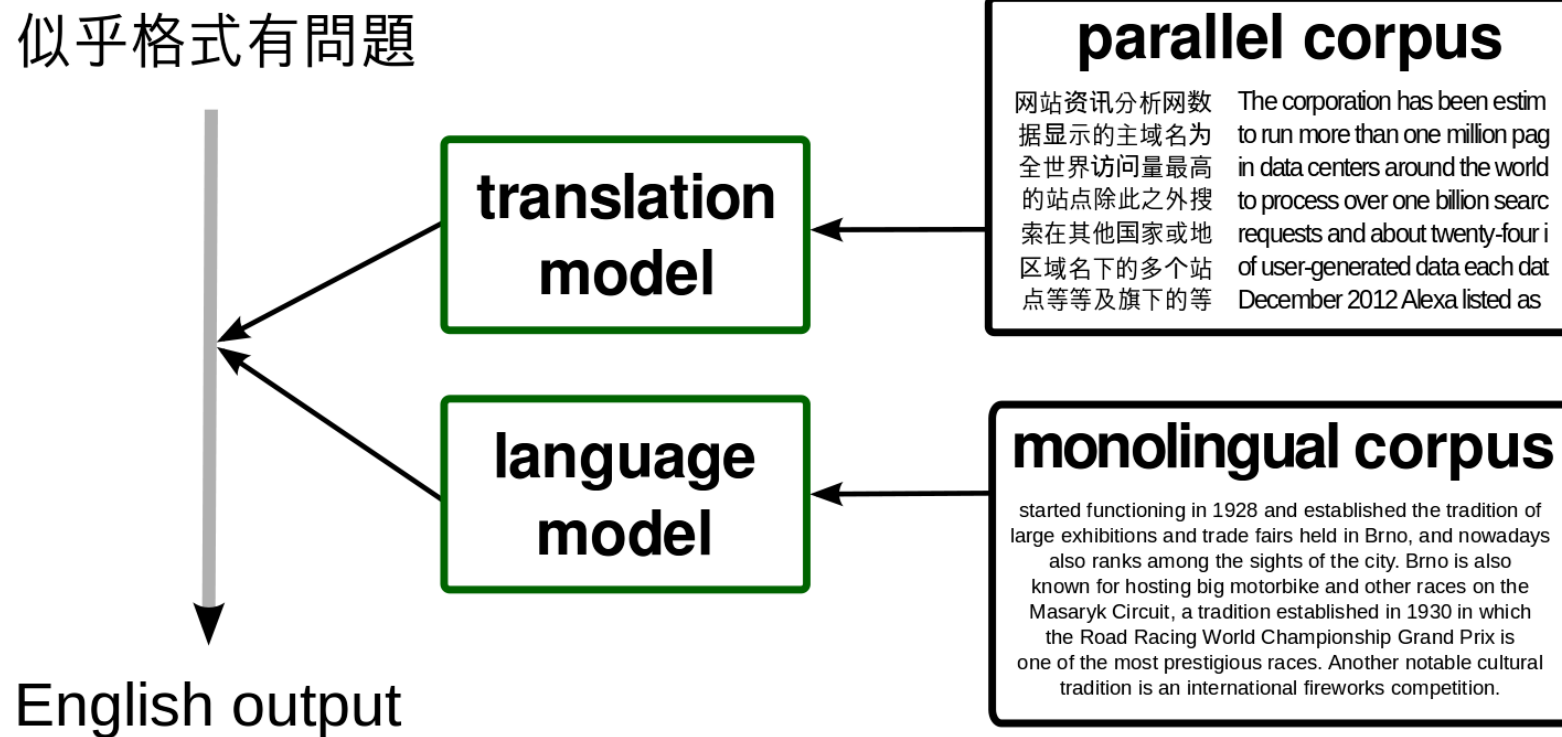
Speech
recognition

Anytime a linguist leaves the group the (speech) recognition rate goes up

- Fred Jelinek

Statistical Machine Translation

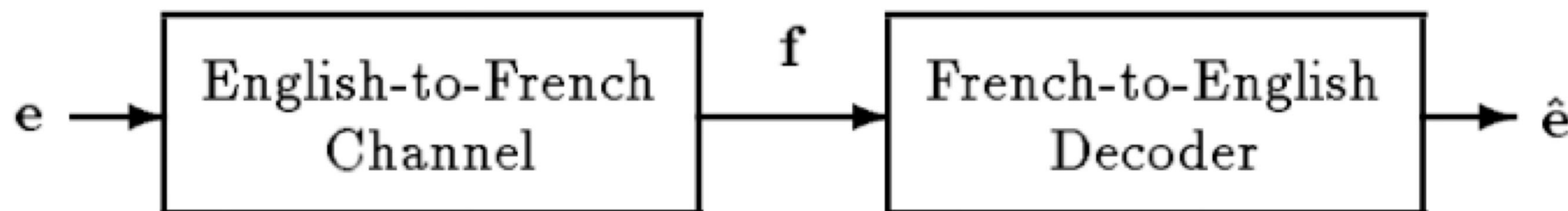
- Required a parallel corpora



The Candide System for Machine Translation

Adam L. Berger, Peter F. Brown, Stephen A. Della Pietra, Vincent J. Della Pietra,
John R. Gillett, John D. Lafferty, Robert L. Mercer*, Harry Printz, Luboš Ureš*

IBM Thomas J. Watson Research Center
P.O. Box 704
Yorktown Heights, NY 10598



$$\hat{e} = \operatorname{argmax}_e \Pr(e \mid f) = \operatorname{argmax}_e \Pr(f \mid e) \Pr(e)$$

Bayes Rule

Rise of statistical learning

- Required human-designed representations and input features

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(64) = 4.15$

Can these features be automatically learned?

Promise of deep learning

- Most NLP works in the past focused on human-designed representations and input features

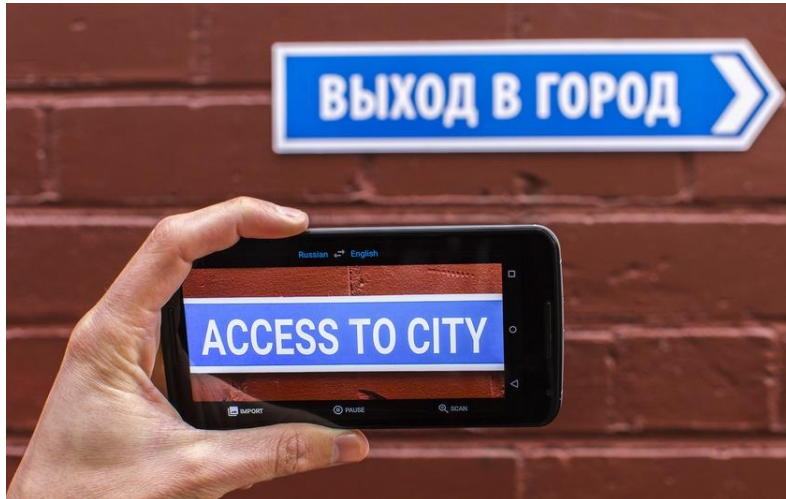
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x_6	log(word count of doc)	$\ln(64) = 4.15$

- **Representation learning attempts to automatically learn good features and representations**
- **Deep learning attempts to learn multiple levels of representation on increasing complexity/abstraction**

Deep learning era

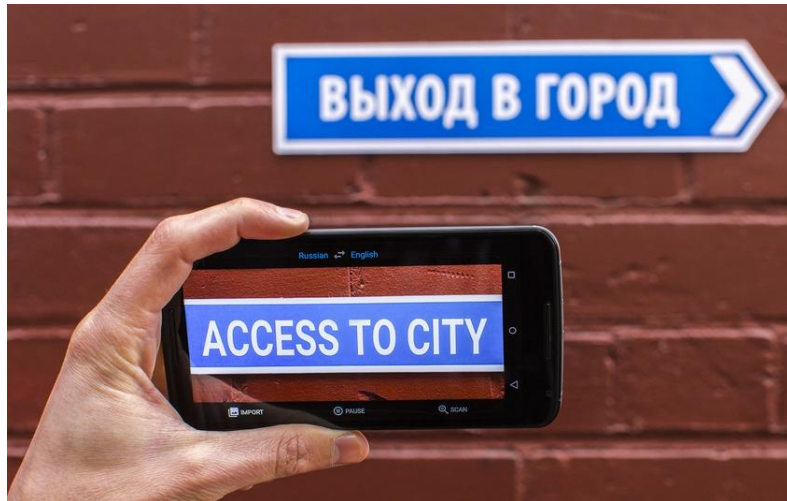
2014-Present: Neural Machine Translation

- Led to significant advances in core NLP technologies



Deep learning era

- Significant advances in core NLP technologies
- **Essential ingredient:** large-scale supervision, lots of compute
- Reduced manual effort - less/zero feature engineering



36M sentence pairs

Russian: Машинный перевод - это круто!



English: Machine translation is cool!

Machine Translation

BBC

Sign in

选项 (英文)

检索

NEWS | 中文

繁

主页 | 国际 | 两岸 | 英国 | 评论 | 科技 | 财经 | 图辑 | 音频材料 | 视频材料 | BBC英伦网

巴拿马首任驻华大使专访：与台湾断交之后

他说，不认同中国“买走”台湾邦交国的说法，与中国建立关系对巴拿马有利，不担心影响与美国关系。

🕒 1小时前

巴拿马外交转向周年 中美大国博弈内幕解密

尼加拉瓜运河成谜：人走楼空 “不再提及”

触发萨尔瓦多与台湾断交的港口令美国担忧



特别推荐



人民币贬值、楼价高涨：消费降级中产失去的优质生活

广告



观点：高铁“一地两检”——中国的强势 港人的无力

江沂：两地价值观和制度差异巨大，冲突无可避免。香港没有反对的权利，也没有反对的能力。



范冰冰消失百日后 中国娱乐业的寒蝉效应

日前发布的《中国影视明星社会责任研究报告》中，最高分徐峥为78分，最低分范冰冰为0分。

🕒 2小时前

中蒙参加俄罗斯军演 “中俄靠拢论”再吸眼球

🕒 2018年 9月 11日

日本人脚踹慰安妇铜像引爆台湾人抗议

🕒 2018年 9月 11日

太空望远镜探测比“三



台湾民间发起东京奥运“正名”公投的意义



Machine Translation

 Sign in

Option (English)

检索

NEWS | 中文

Traditional

Homepage | International | Cross-strait | United Kingdom | comment | Technology | Finance | Picture series | Audio material

Interview with Panama's first ambassador to China: After breaking diplomatic relations with Taiwan

He said that he does not agree with China's saying that "buy" Taiwan's diplomatic relations with China. Establishing relations with China is beneficial to Panama and does not worry about affecting relations with the United States.

1 hour ago

Panamanian Diplomacy Turns to Anniversary

The Nicaragua Canal is a mystery: people go to the floor and "no longer mention"

The port that triggered the break of El Salvador and Taiwan has worried the United States



Special recommendation

Renminbi depreciation, property prices are rising: consumption downgrades the loss of quality life in the middle class

ADVERTISING



The significance of the Taiwanese people's "referred to" referendum



China and Mongolia participate in the Russian military

Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

<https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html>

<i>Input sentence:</i>	<i>Translation (PBMT):</i>	<i>Translation (GNMT):</i>	<i>Translation (human):</i>
李克強此行將啟動中加總理年度對話機制，與加拿大總理杜魯多舉行兩國總理首次年度對話。	Li Keqiang premier added this line to start the annual dialogue mechanism with the Canadian Prime Minister Trudeau two prime ministers held its first annual session.	Li Keqiang will start the annual dialogue mechanism with Prime Minister Trudeau of Canada and hold the first annual dialogue between the two premiers.	Li Keqiang will initiate the annual dialogue mechanism between premiers of China and Canada during this visit, and hold the first annual dialogue with Premier Trudeau of Canada.

(Wu et al., 2016)

Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

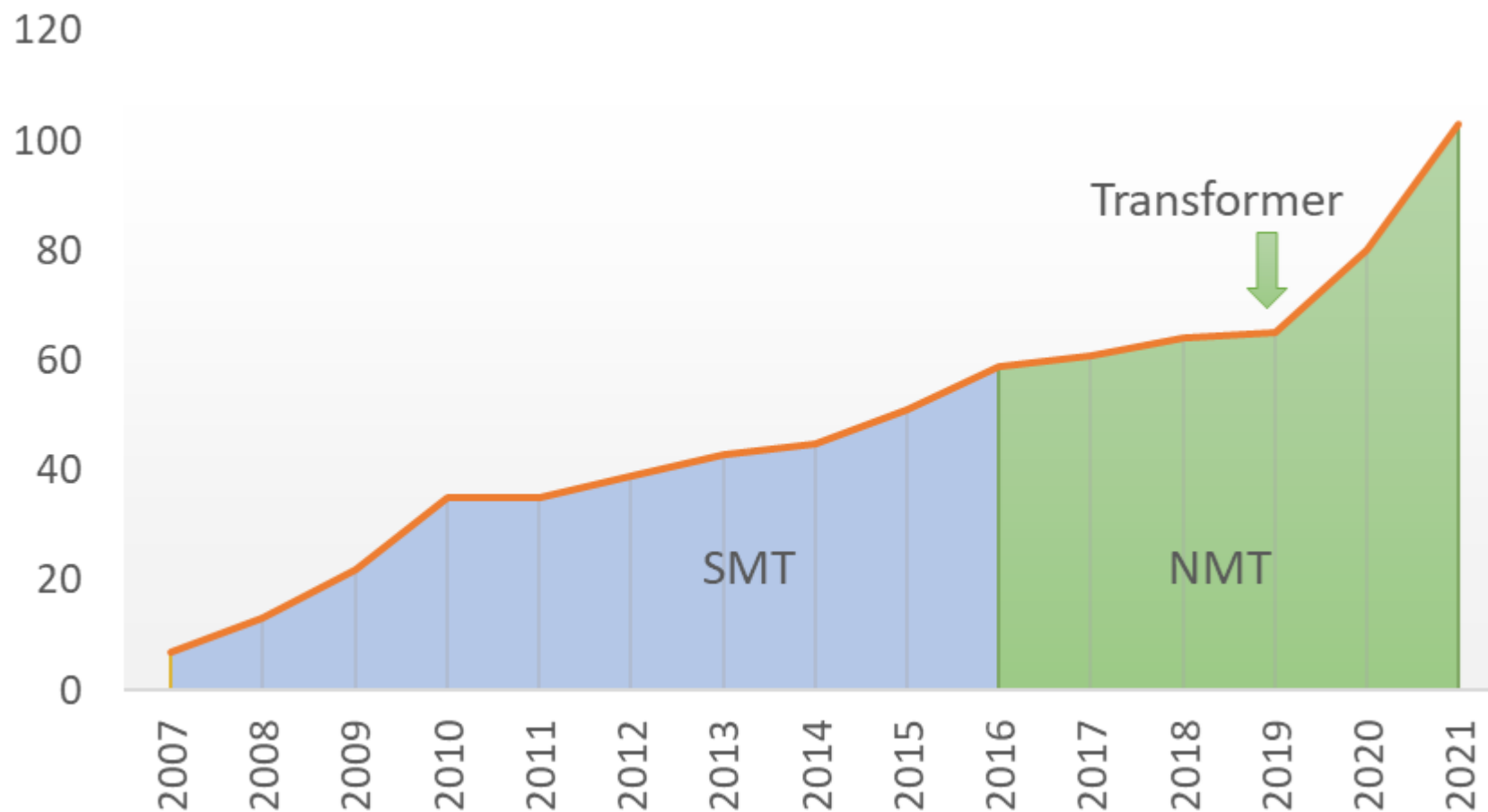
<https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html>

Table 10: Mean of side-by-side scores on production data

	PBMT	GNMT	Human	Relative Improvement
English → Spanish	4.885	5.428	5.504	87%
English → French	4.932	5.295	5.496	64%
English → Chinese	4.035	4.594	4.987	58%
Spanish → English	4.872	5.187	5.372	63%
French → English	5.046	5.343	5.404	83%
Chinese → English	3.694	4.263	4.636	60%

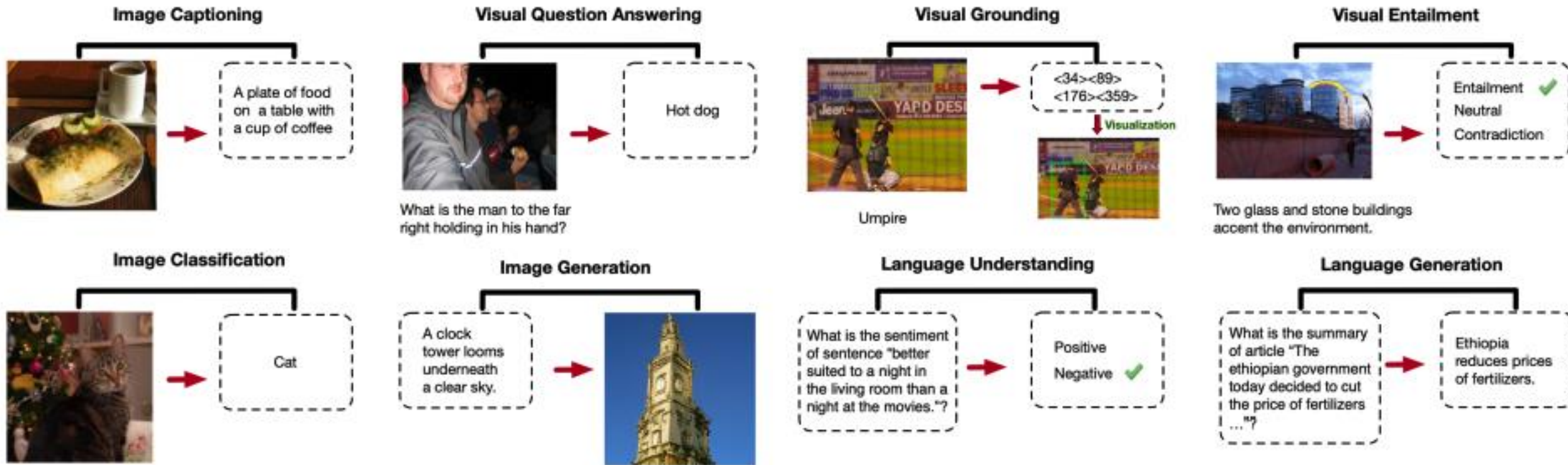
(Wu et al., 2016)

Machine Translation (across 100 languages)



<https://www.microsoft.com/en-us/research/blog/microsoft-translator-now-translating-100-languages-and-counting>

Multimodal models



<https://github.com/OFA-Sys/OFA>

OFA (Wang et al, 2022)

Power of Data

CleverBot (2010)

User:	Do you like Doctor Who?
Cleverbot:	Yes, so funny.
User:	What's your favorite show?
Cleverbot:	I don't have one.
User:	What's your name?
Cleverbot:	Jason I already told you.
User:	No your real name.
Cleverbot:	Stephanie IS my real name.
User:	I thought it was Jason?
Cleverbot:	How did you come to that conclusion.
User:	Because you said so.
Cleverbot:	I have to do my homework.

<https://www.cleverbot.com/>

How it works:

- Corpus of conversational turns
- Find the most similar sentence and copy the response
- Learn from human input

What do you get?

- Something that someone say
- Incoherent conversation

Power of Data

Meena (Google, 2020)

Conversation D

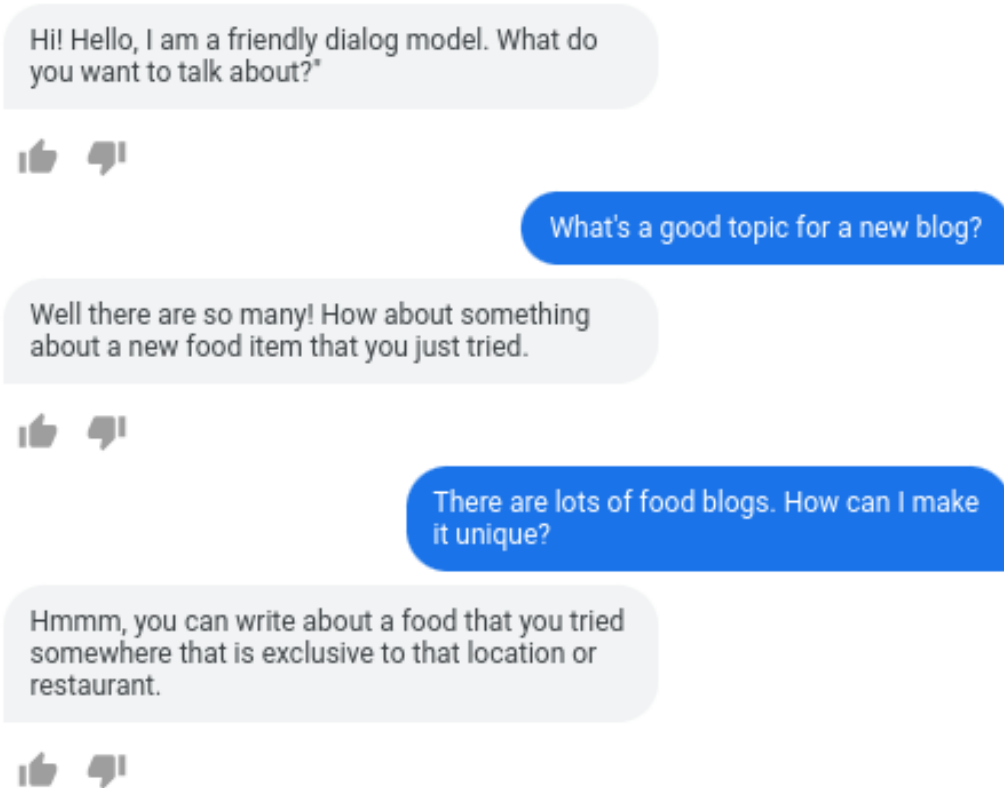
Human: *I like how cute cows are*
Meena: *Ah, interesting. You like how they are cute, right?*
Human: *yeah, and they're really smart*
Meena: *I heard they go to college*
Human: *cows go to college?*
Meena: *I heard that a cow went to Harvard.*
Human: *what did the cow study?*
Meena: *Bovine sciences.*
Human: *do horses go to Harvard?*
Meena: *Horses go to Hayvard.*
Human: *that's a pretty good joke, I feel like you led me into it*
Meena: *You were trying to steer it elsewhere, I can see it.*

How it works:

- Corpus of conversational turns (over 40B words)
- Train huge neural network (2.6 billion parameters) for 30 days on 2048 TPUs cores
- Predict response given a sentence

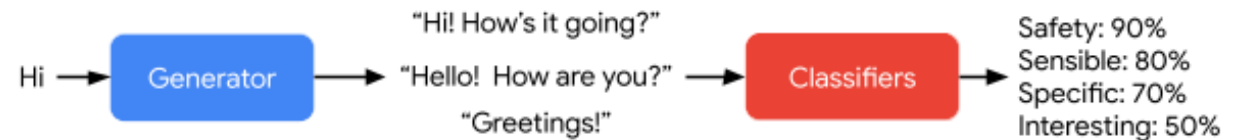
Power of Data

LaMDA (Google, 2022)



How it works:

- Pretrain on even more data
- Fine-tune so it has certain characteristics

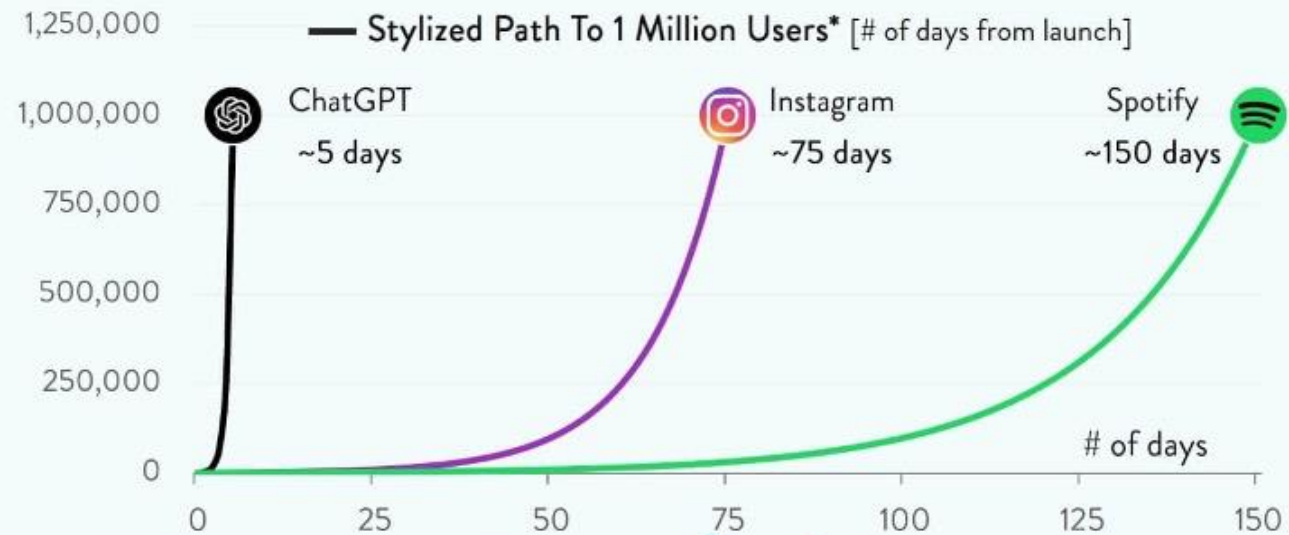
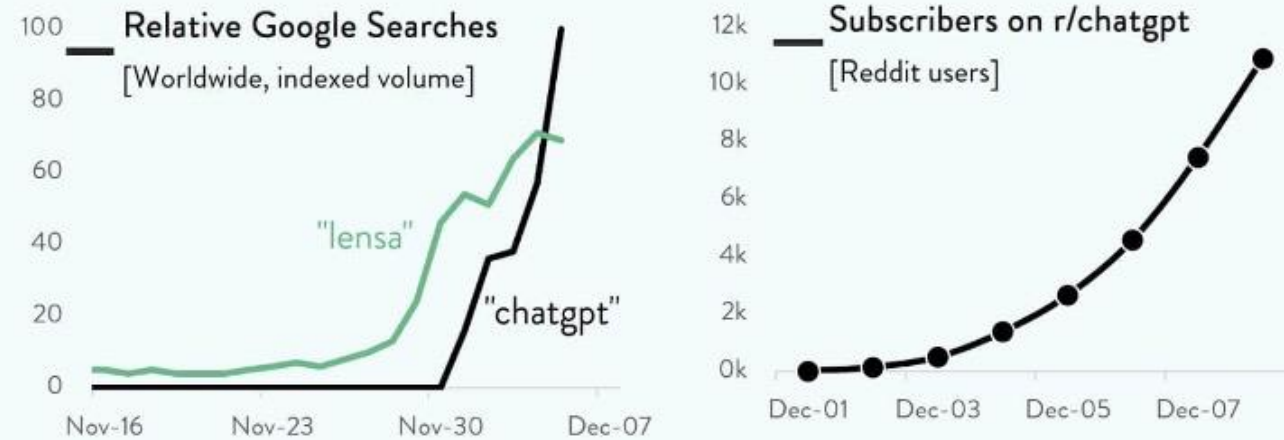


Is this chatbot sentient?

<https://ai.googleblog.com/2022/01/lamda-towards-safe-grounded-and-high.html>

https://youtu.be/CvD9U_3hBPo

ChatGPT From OpenAI Is A Bot Taking The Tech World By Storm



Sources: Google, Subredditstats, Media Reports

chart*Path is stylized to the 1m milestone

Source: Chartr

Power of Data

ChatGPT (OpenAI, 2022)

<https://openai.com/blog/chatgpt/>



write a sonnet about pokemon



A world of wonder, filled with creatures wild,
A realm of magic, where adventures start,
From Kanto to Kalos, far and wide,
The world of Pokémon, alive and heart.

With Pikachu as guide, we set out to roam,
To catch and train them all, to be the best,
Battling Gym Leaders, overcoming foes,
Our bond with creatures strong, a bond we'll never test.

But it's not just battles that define,
The world of Pokémon, so full of cheer,
It's friendships formed, and lessons learned,
A world that's full of love, and never fear.

So let's go on a journey, you and I,
Through the world of Pokémon, beneath the sky.

How it works:

- Trained on conversational data
- Uses reinforcement learning with human feedback

The Brilliance and Weirdness of ChatGPT

A new chatbot from OpenAI is inspiring awe, fear, stunts and attempts to circumvent its guardrails.

<https://www.nytimes.com/2022/12/05/technology/chatgpt-ai-twitter.html>

Time taken to reach 1m users (mths)

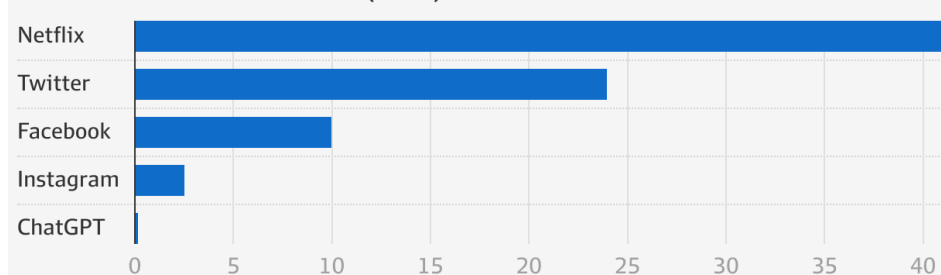


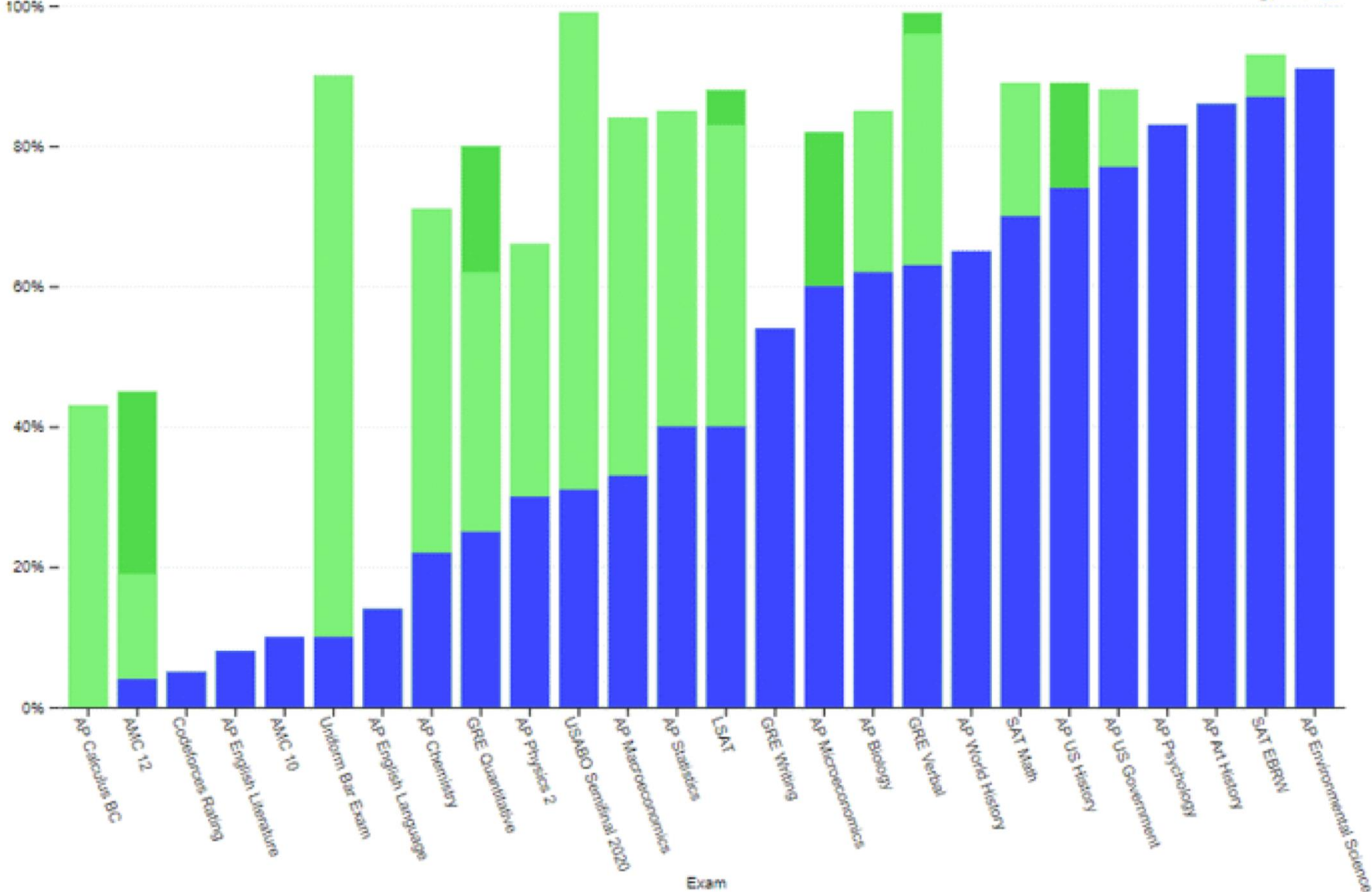
Chart: Financial Review • Source: Genevieve Roch-Decter, CFA

Exam results (ordered by GPT-3.5 performance)

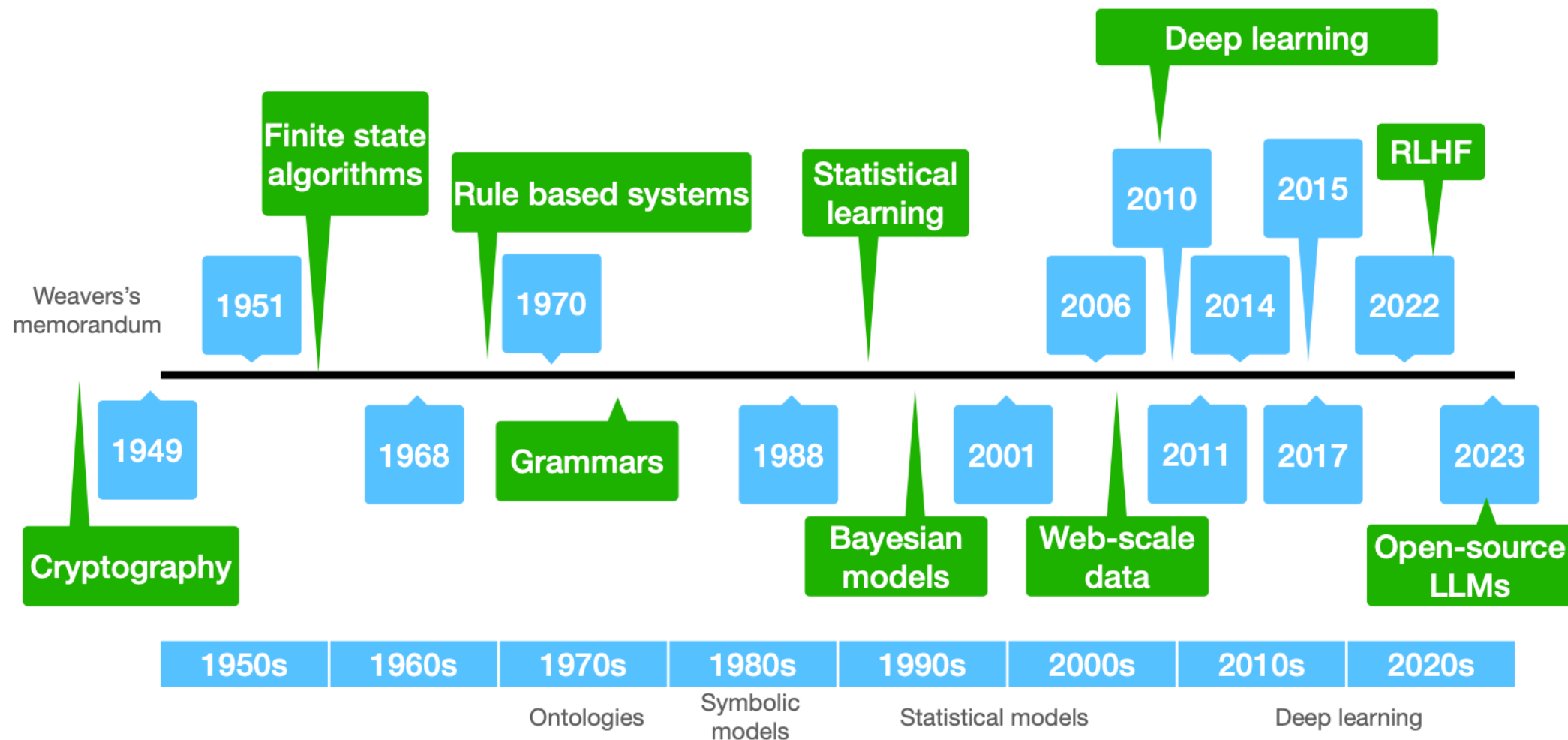
Estimated percentile lower bound (among test takers)

100% —

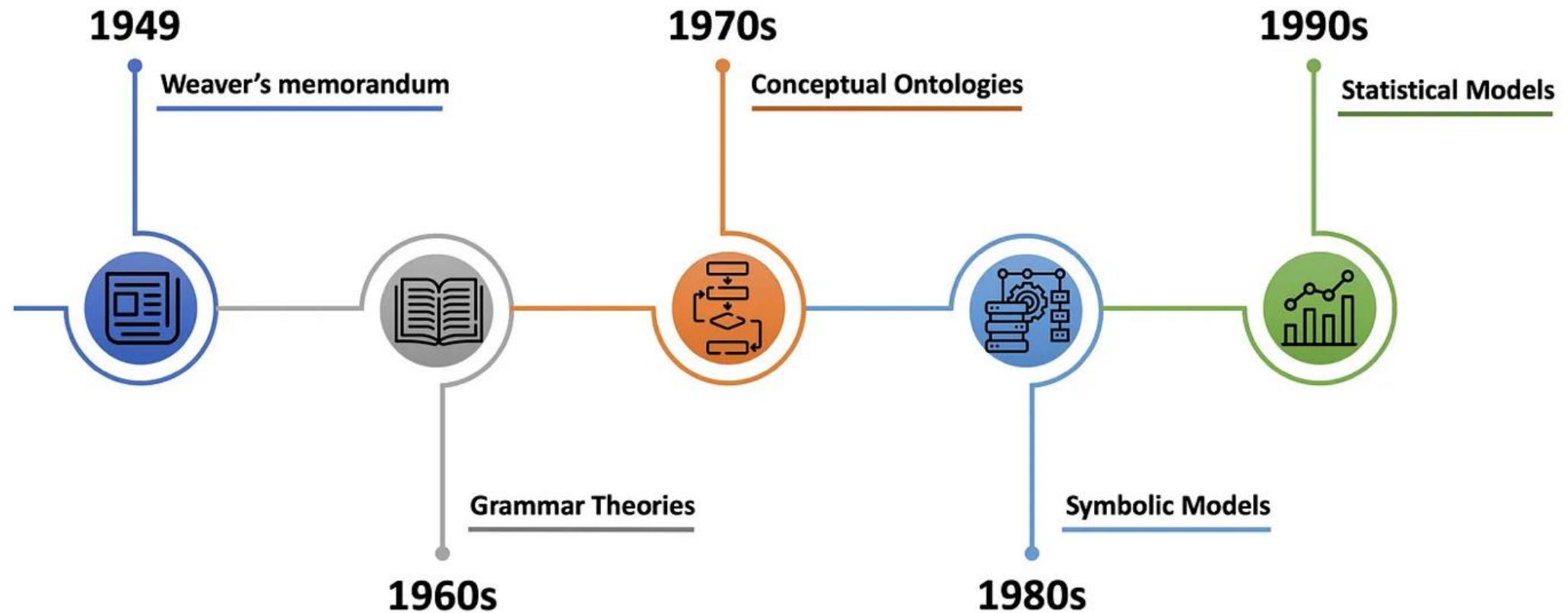
gpt-4
gpt-4 (no vision)
gpt-3.5



Key developments in NLP

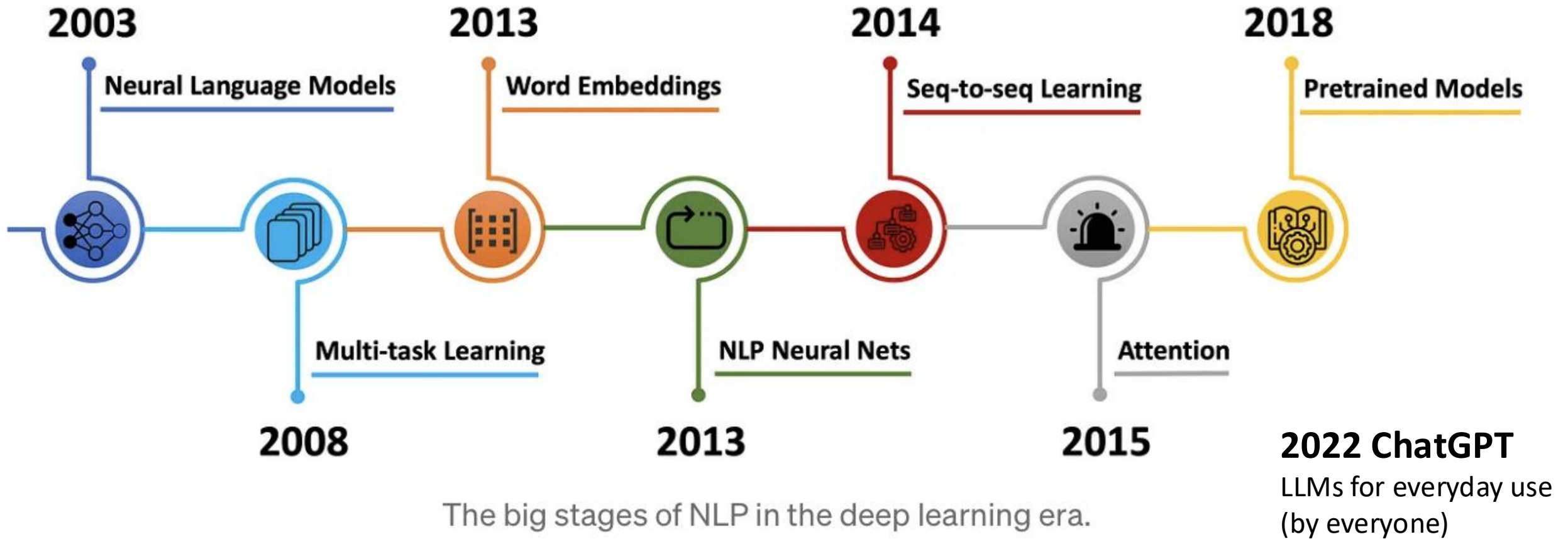


NLP before deep learning



The big stages of NLP before the deep learning era.

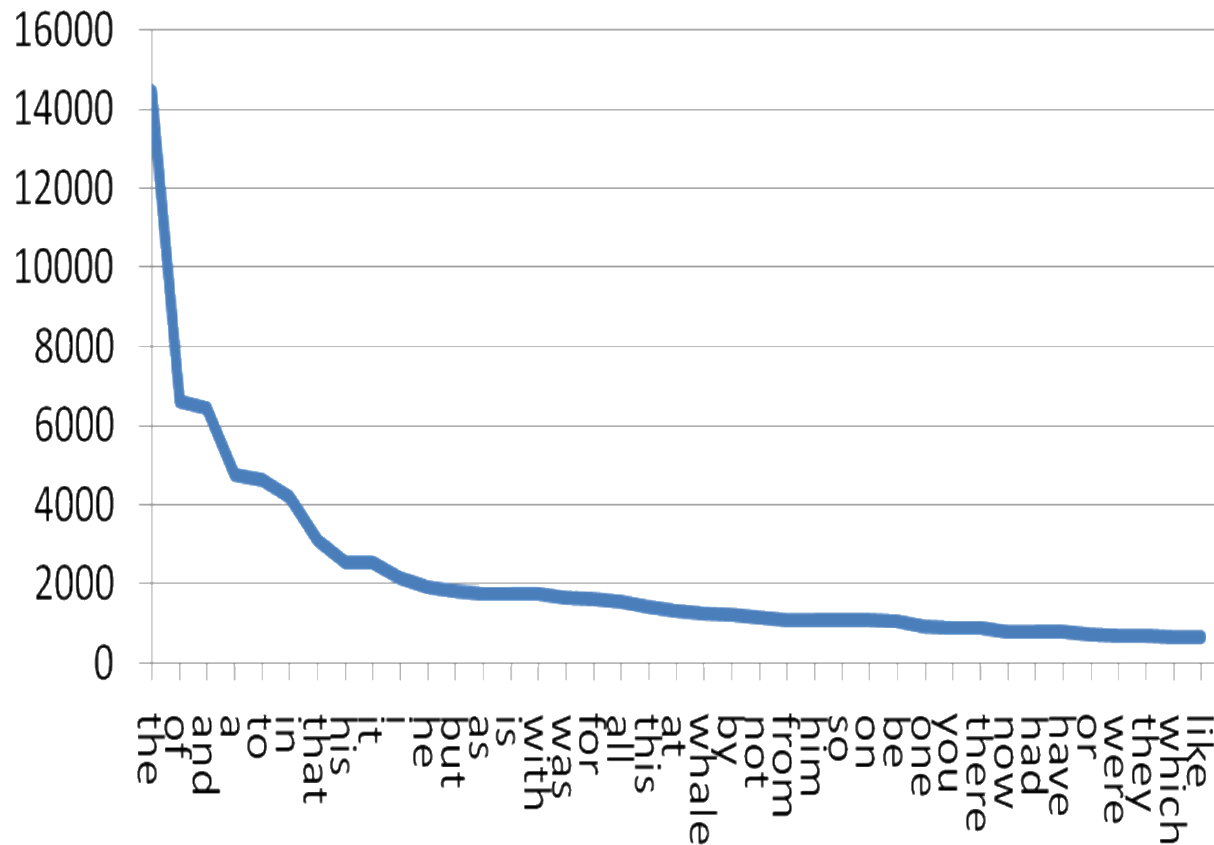
NLP in the deep learning era



Remaining challenges in NLP?

Challenges in modern NLP

- *Scale*: Large number of phenomena
- *Sparsity*: Text data is often heavy-tailed



- *Solved by ingesting the Internet?*
- *But what about low-resource languages? Specialized domains?*

Ambiguity in grounding

I saw her duck



Challenges in modern NLP

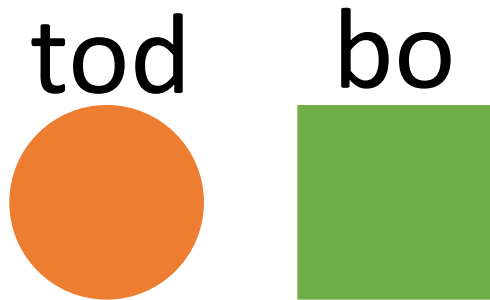
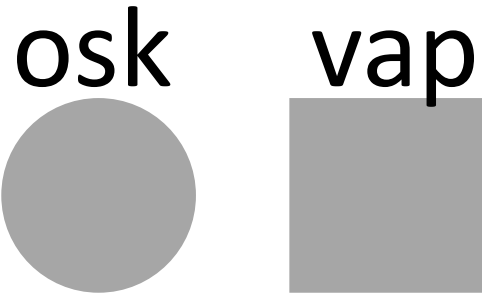
- *Bias*: Models learn biases in available data



- *Context*: Knowledge bases, perception, interaction

Compositional generalization

Grounding



Compositionality



Generalization

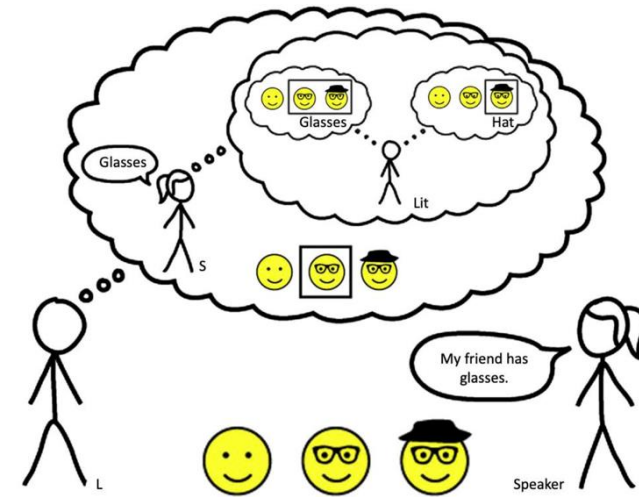


Challenges of understanding language

- Does solving language require solving AI?
- Language derives its meaning from use



'block' 'pillar' 'slab' 'beam'.



- To truly understand language: need **context**, **world knowledge**, **mental model** of other people

Grounded natural language

Move **OBJECT** from **START** to **GOAL**.

Move *the stuffed animal* from *the chair* to *the sofa*.



*Move toward
multimodal models and
action taking agents*



HomeRobot: Open Vocabulary Mobile Manipulation

[Yenamandra et al. CoRL 2023]

<https://ovmm.github.io>

What will this course cover?

What will this course cover?

Representations

- Embeddings
- Compositional
- Structured

Methods

- Language Models
- Statistical Methods
- Neural Models
- Dynamic programming

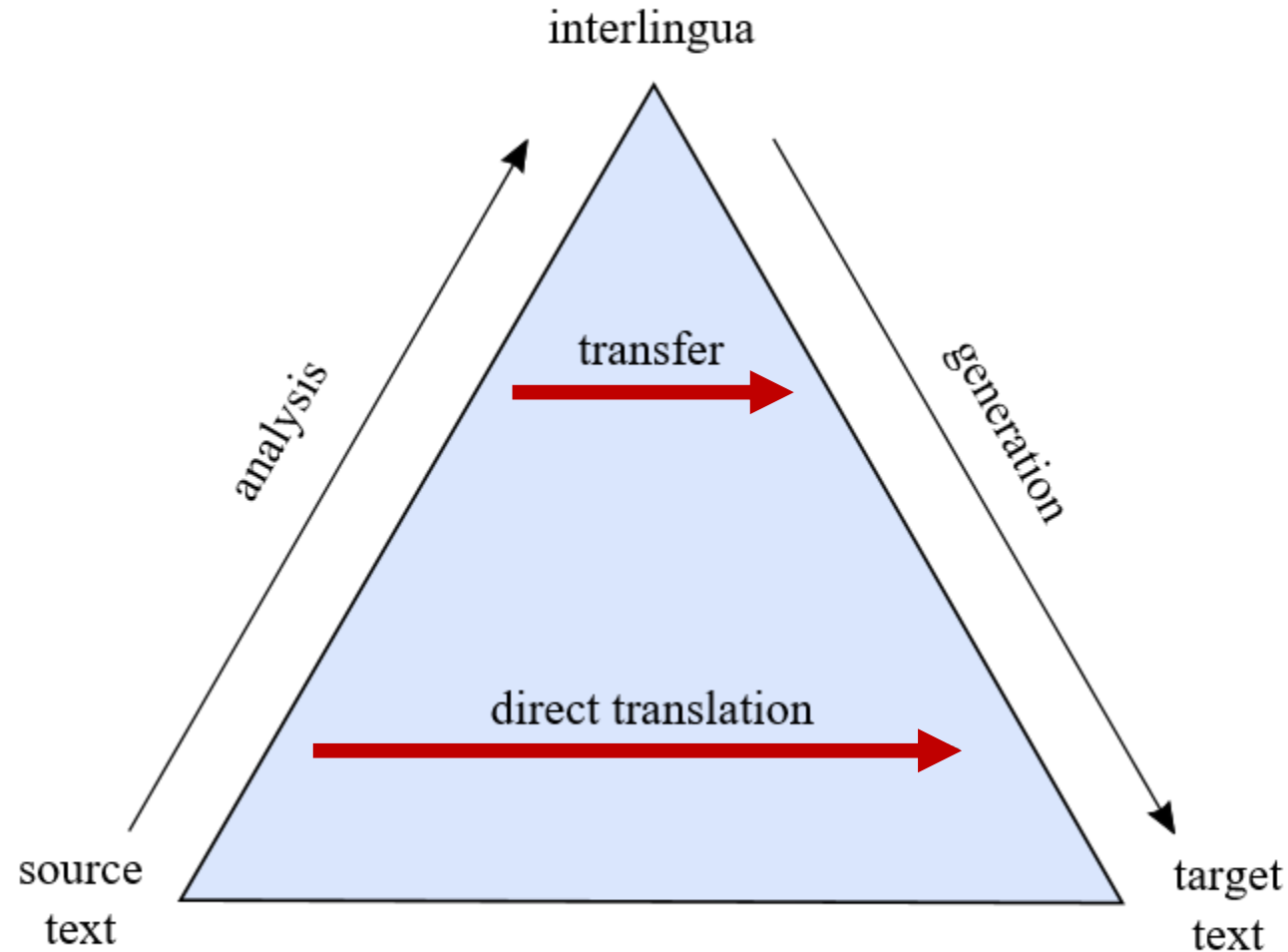
LLMs

- Pretrained models
- Post-training
- Prompting
- Efficient fine-tuning
- LLM agents

Applications and tasks

- Machine translation, text generation, question answering, grounding

Vauquois Triangle for translation



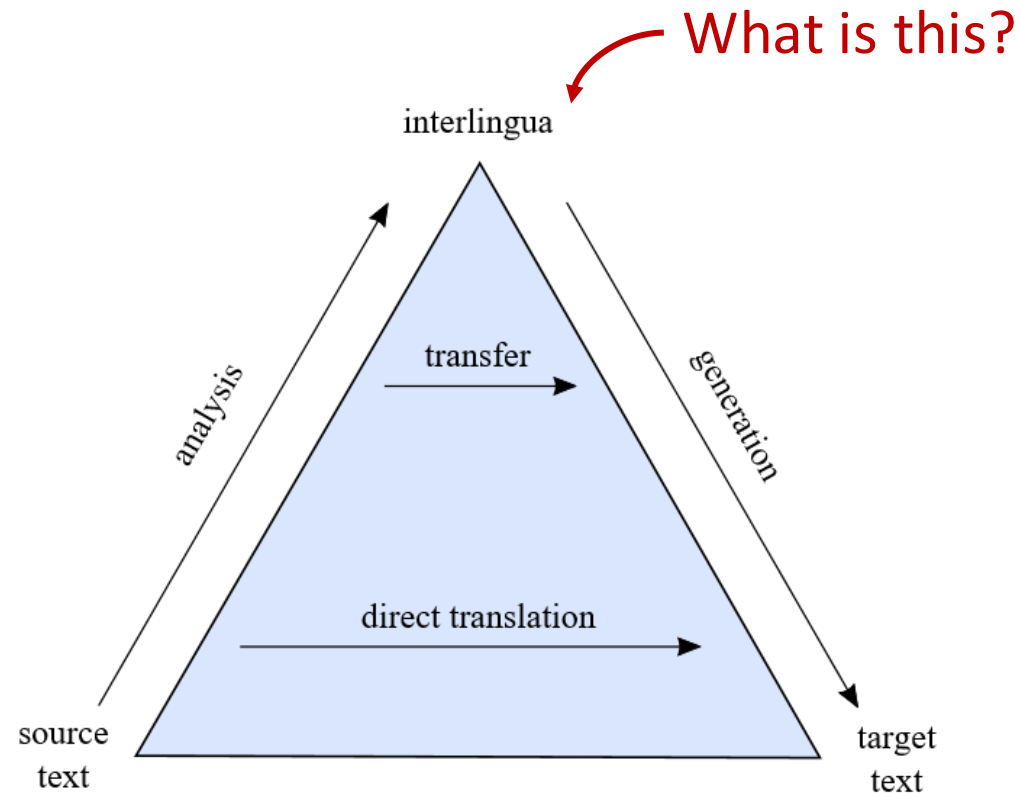
the cat sat on the table

η γάτα κάθισε στο τραπέζι

Key Questions

- How to **represent** language?
- How to **compose** meaning?
- How do we translate from the “surface form” of language to a representation that a computer can process, and back?
 - How do we **encode** / **parse** the meaning of a sentence?
 - How do we **decode** / **generate** text from a meaning representation?
- What are the key methods / algorithms for **learning** these representations?

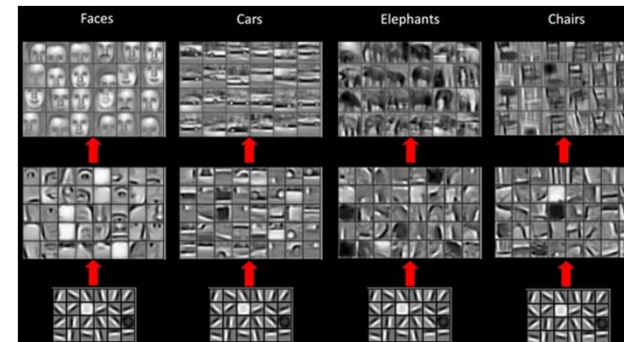
What should the representation look like?



$$L = \begin{bmatrix} \vdots & \vdots & \vdots & \dots & \vdots & \vdots \end{bmatrix}_n$$

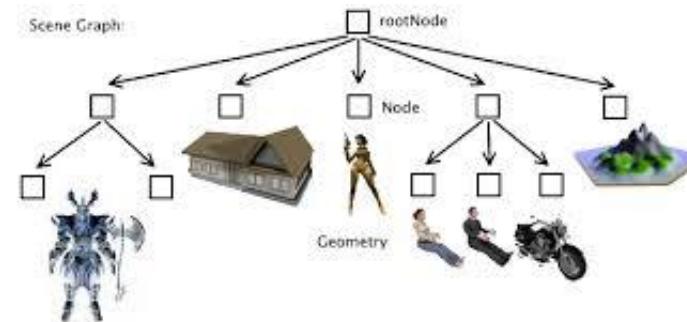
the cat mat ...

$|V|$



Lambda expression:

$((\lambda x . (\lambda y . x)) (\lambda z . z)) (\lambda a . a)$



Representations

How to represent the meaning of something?



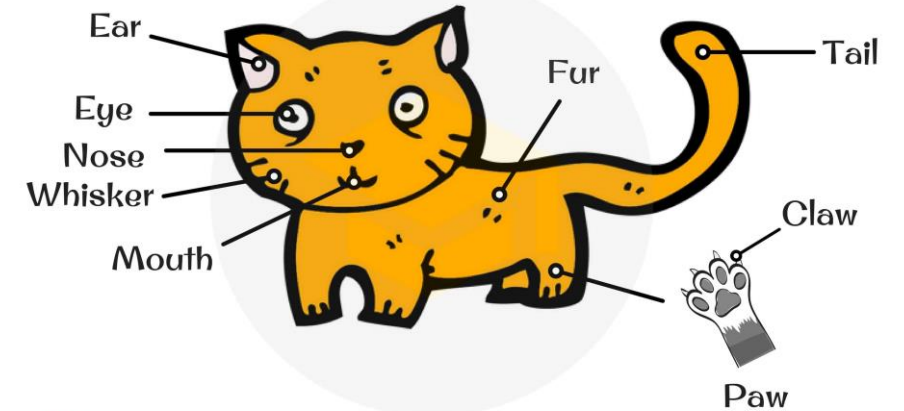
“cat”

cat: a small domesticated carnivore, *Felis domestica* or *F. catus*, bred in a number of varieties.

```
cat → {  
  isMammal: true  
  hasFur: true  
  hasLegs: true  
  meows: true  
  barks: false  
  height: 9.1 – 9.8 in  
  weight: 7.9 – 9.9 lbs  
  ...  
}
```

Attributed
representation

Parts of a cat



TESL.COM

Representations



“cat”



“dog”

Representing meaning as vectors

- common representation space
- enables information sharing
- can be learned from data

- One-hot

cat = [0 0 0 1 0 0 0]

dog = [0 0 0 0 0 1 0]

- Embeddings

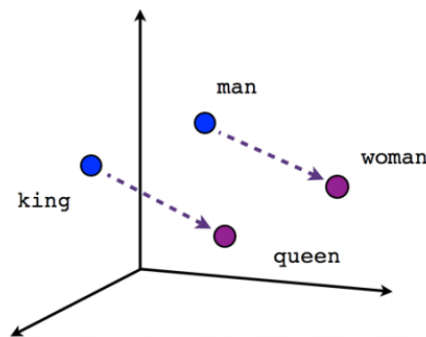
cat = [0.04 1.79 -1.79 1.07 0.48]

dog = [0.61 1.84 -1.12 0.52 0.53]

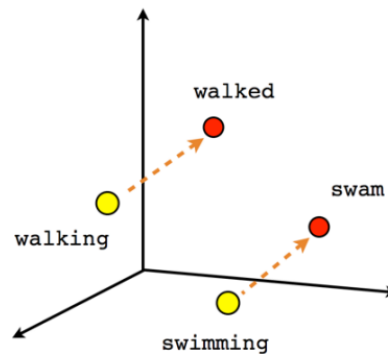
Distributed representations

Project words onto a continuous vector space

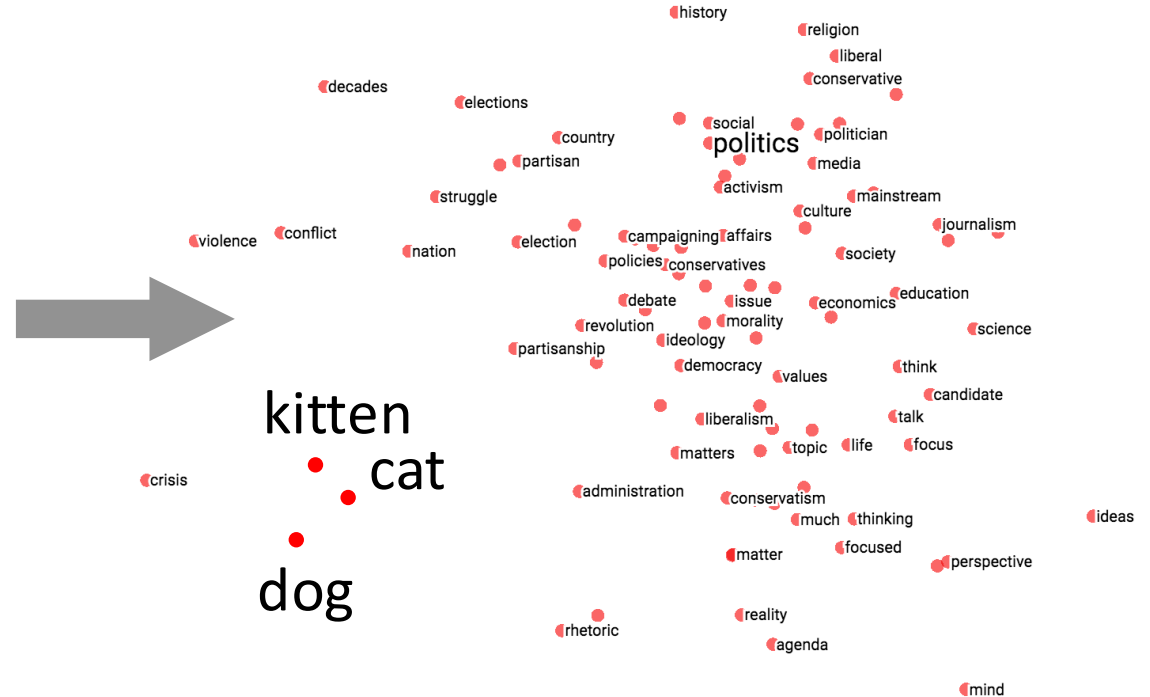
Similar words closer to each other



Male-Female

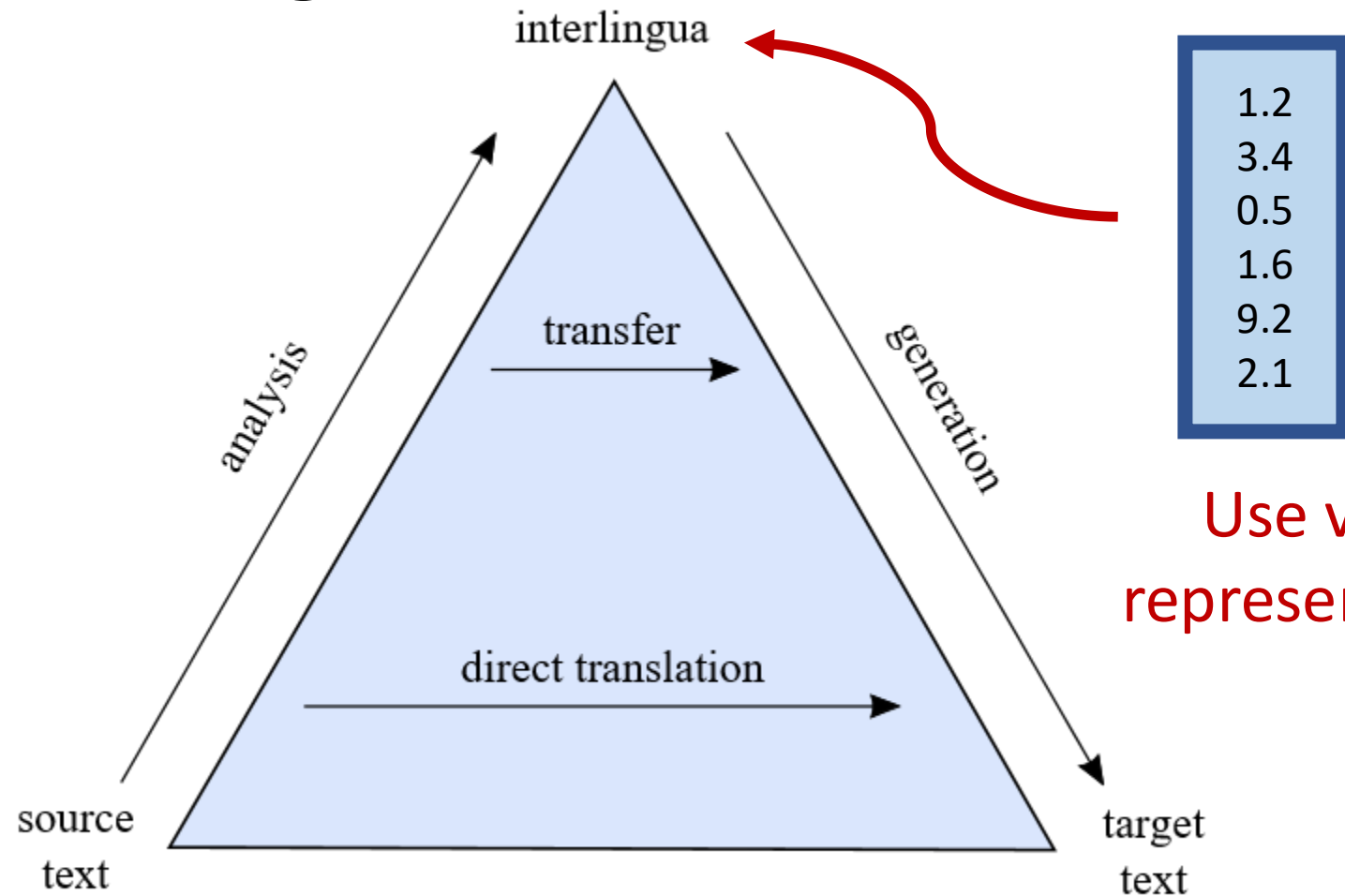


Verb tense



$$v(\text{king}) - v(\text{man}) + v(\text{woman}) = v(\text{queen})$$

Vauquois Triangle for translation

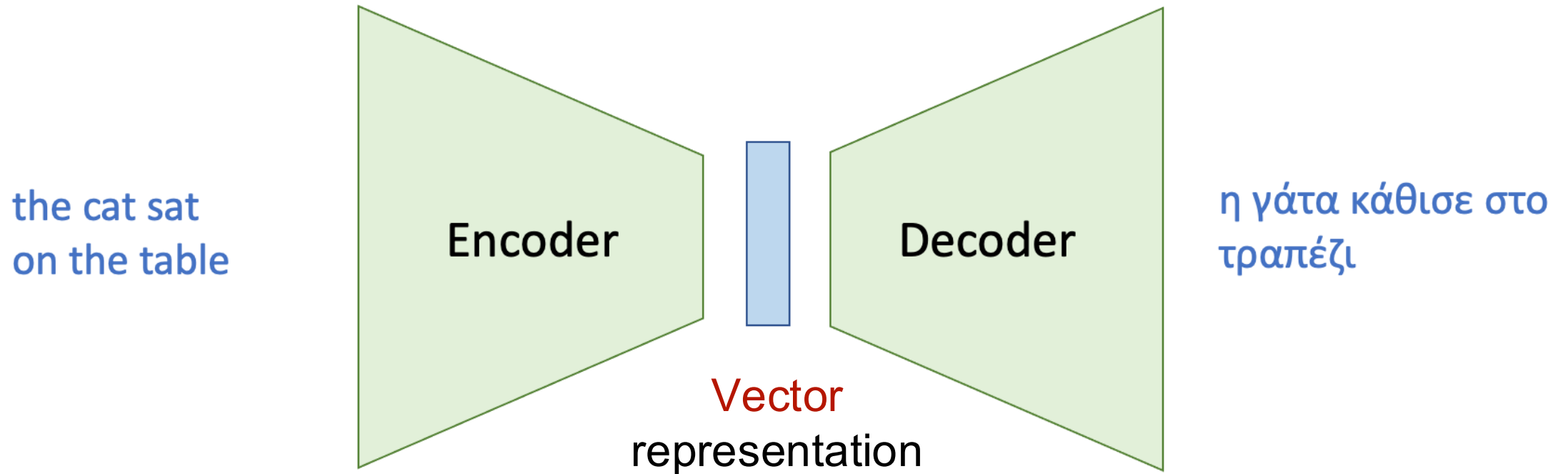


Use vector to
represent meaning

the cat sat on the table

η γάτα κάθισε στο τραπέζι

Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not

Translating across modalities

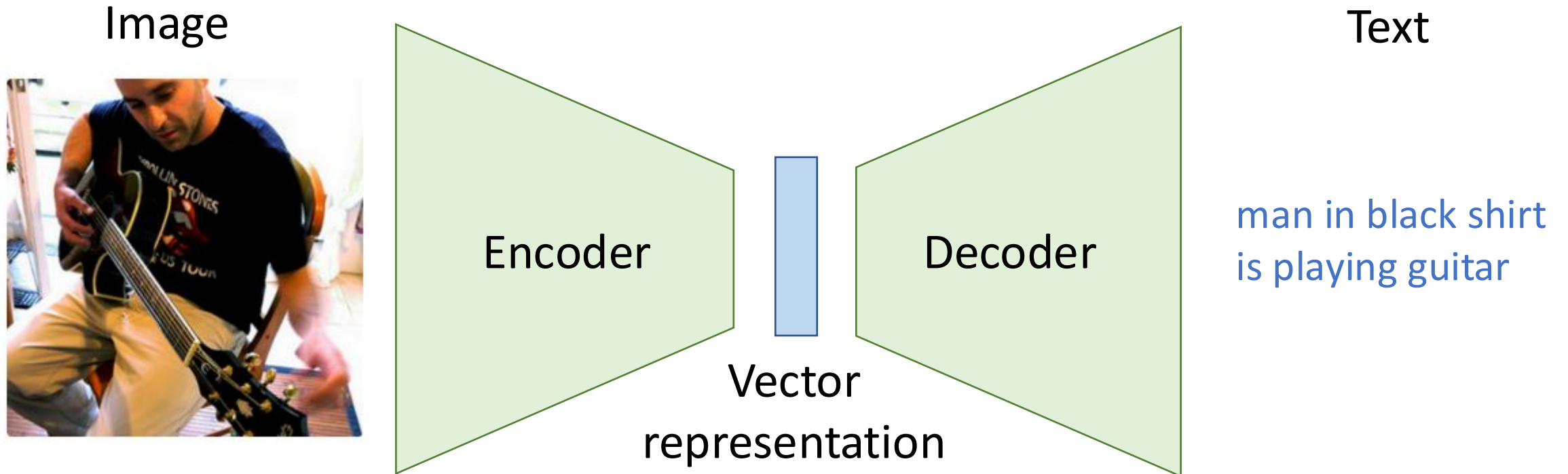
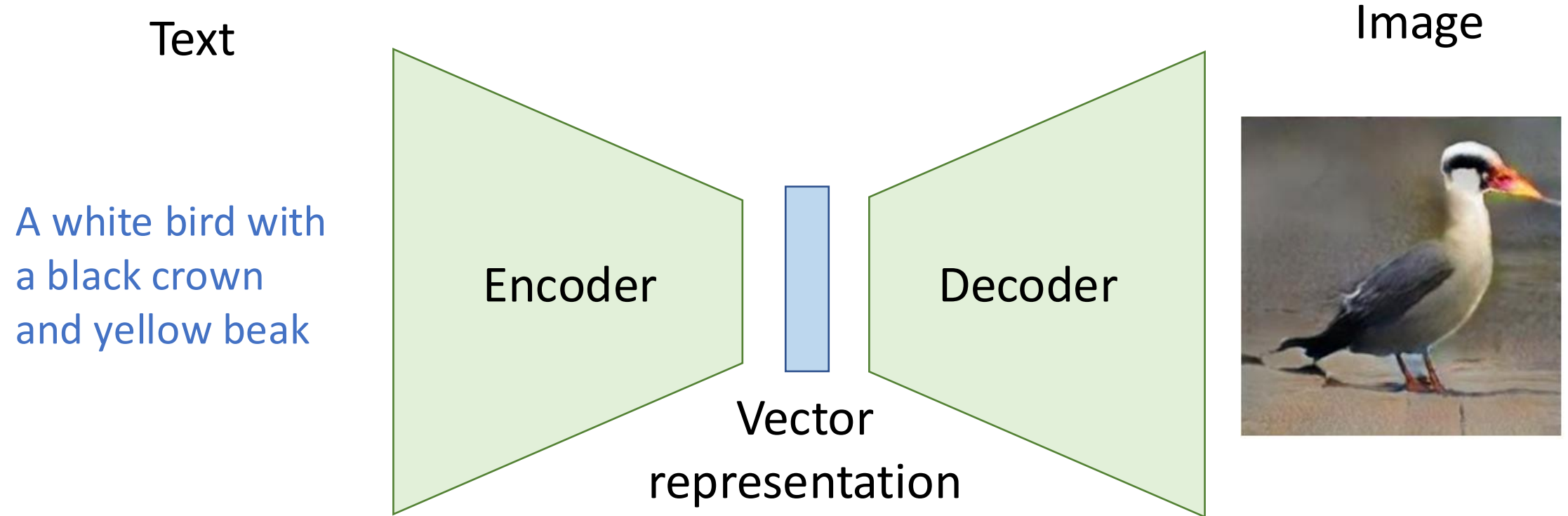


Image captioning

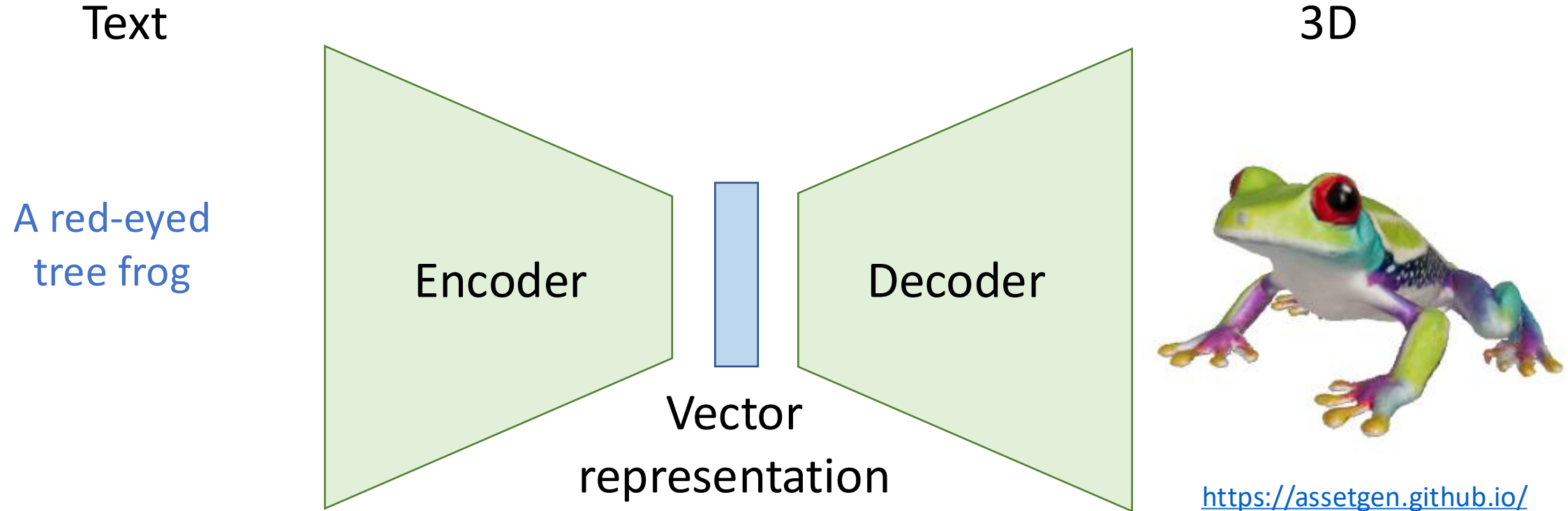
“Deep Visual-Semantic Alignments for Generating Image Descriptions”
[Karpathy and Fei-Fei CVPR 2015]

Translating across modalities

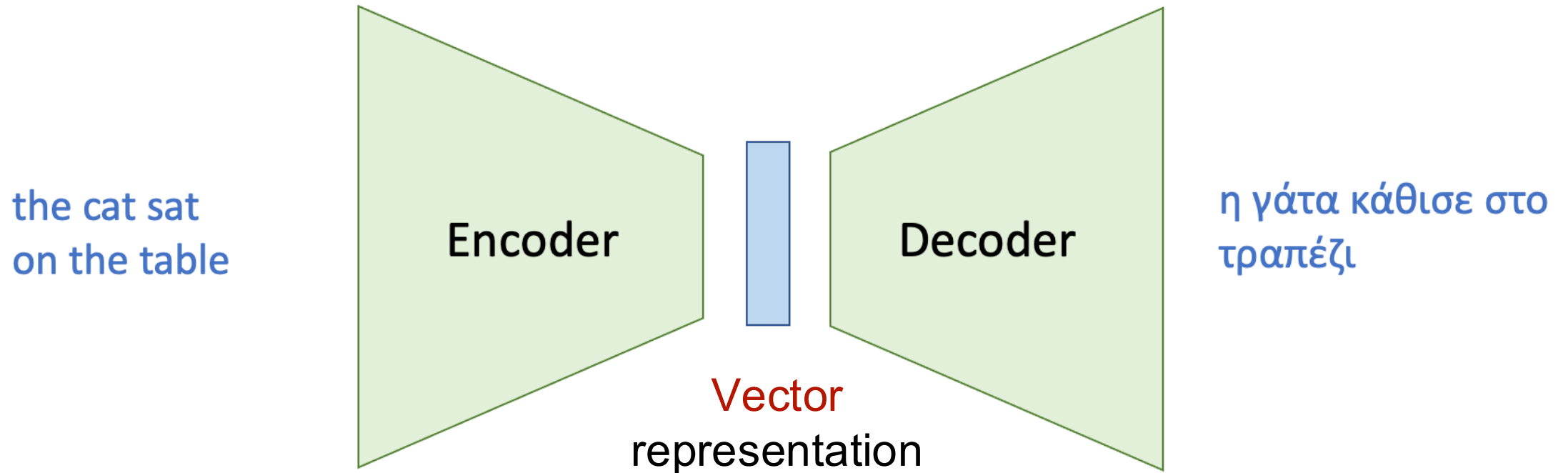


“StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks”
[Zhang et al, ICCV 2017]

Translating across modalities



Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not

How to encode language?

What is this function that takes the sentence and converts it into a vector?

the cat sat
on the table

Encoder

Vector
representation

Encoder

- How to convert the sentence into a meaning representation?

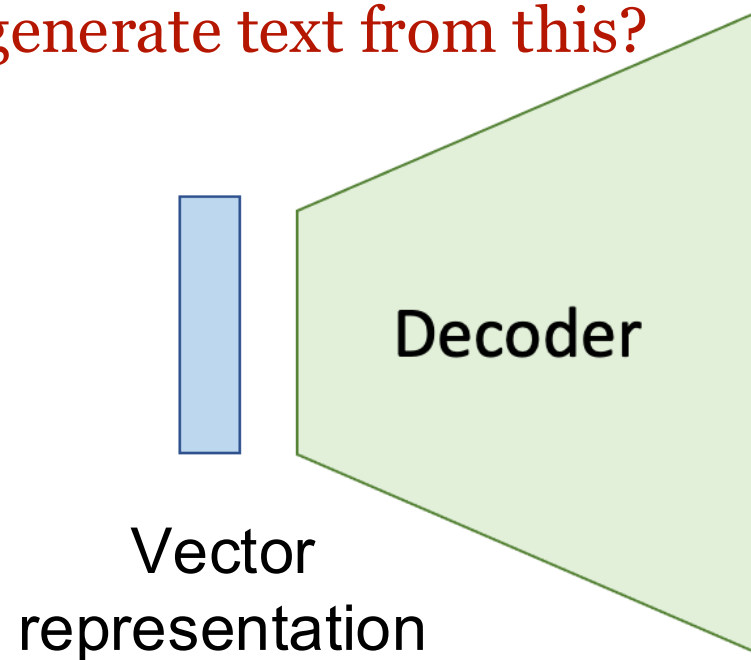
Understanding what is said
(encoding, parsing, feature
extraction)

How to decode language?

How do we generate text from this?

Decoder

- How to take the vector representation and generate a sequence of words?



Deciding what to say
(decoding, generating)

How to encode language?

What is this function that takes the sentence and converts it into a vector?

the cat sat
on the table

Encoder

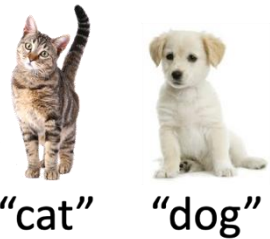
Vector
representation

Properties we want

- Be able to represent a single word
- Be able to represent the meaning of a sentence as a composition of the meaning of each word

Understanding what is said
(encoding, parsing, feature
extraction)

How to learn representations for words?



start with representing words as discrete tokens

- One-hot

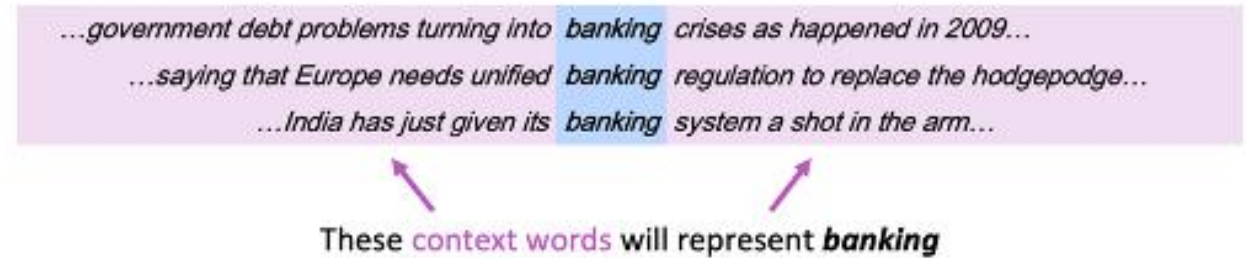
cat = [0 0 0 1 0 0 0]
dog = [0 0 0 0 0 1 0]

learn about neural models that use vectors

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]
dog = [0.61 1.84 -1.12 0.52 0.53]

use **context** to represent words



learn a dense vector representation that can be used to **predict** words from their context

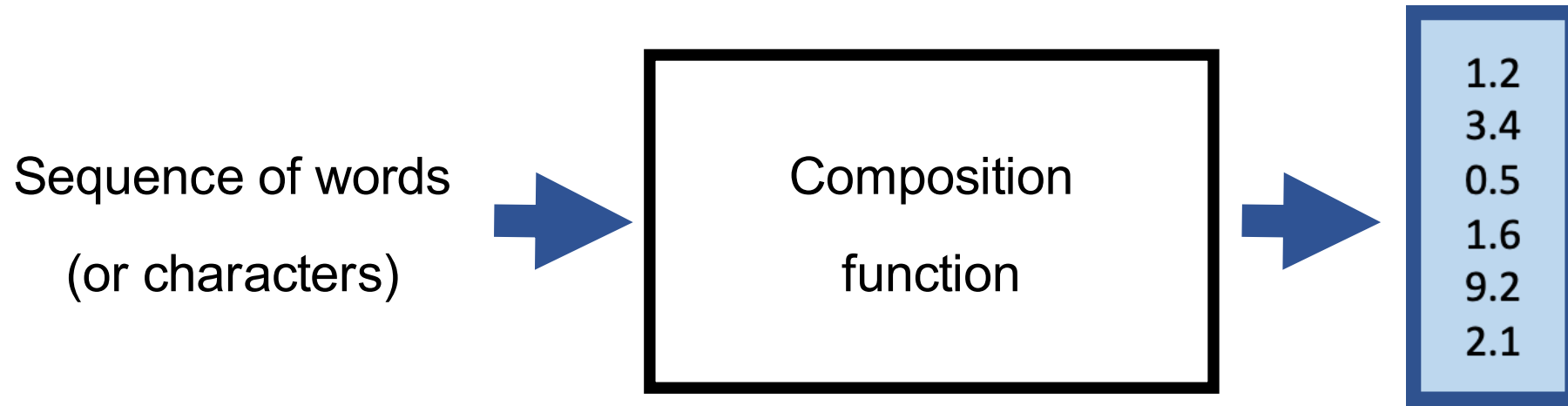
- Predict what goes next

The cat sat on the _____

- Fill in the blank

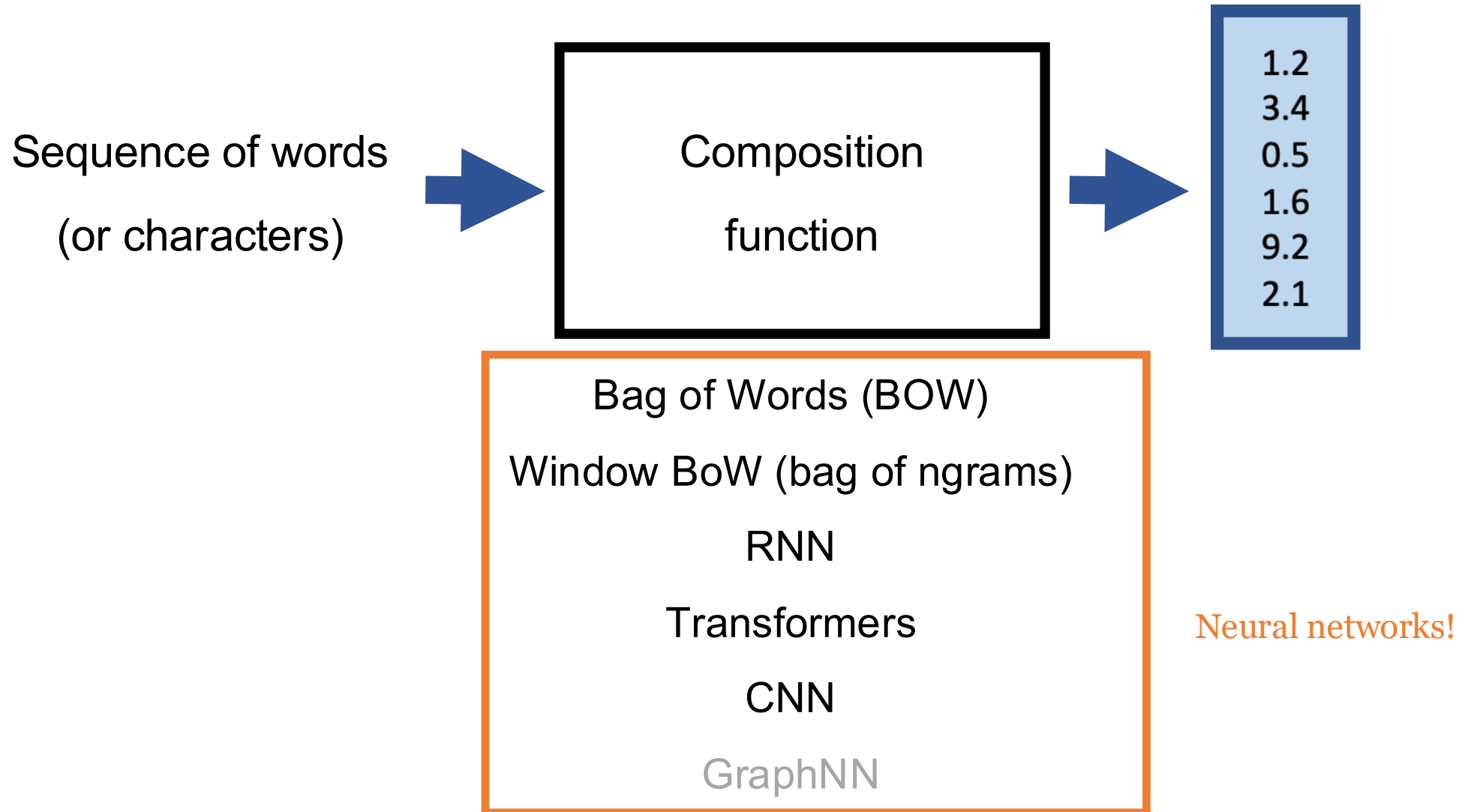
A bottle of _____ is on the table.

How to **compose** language?

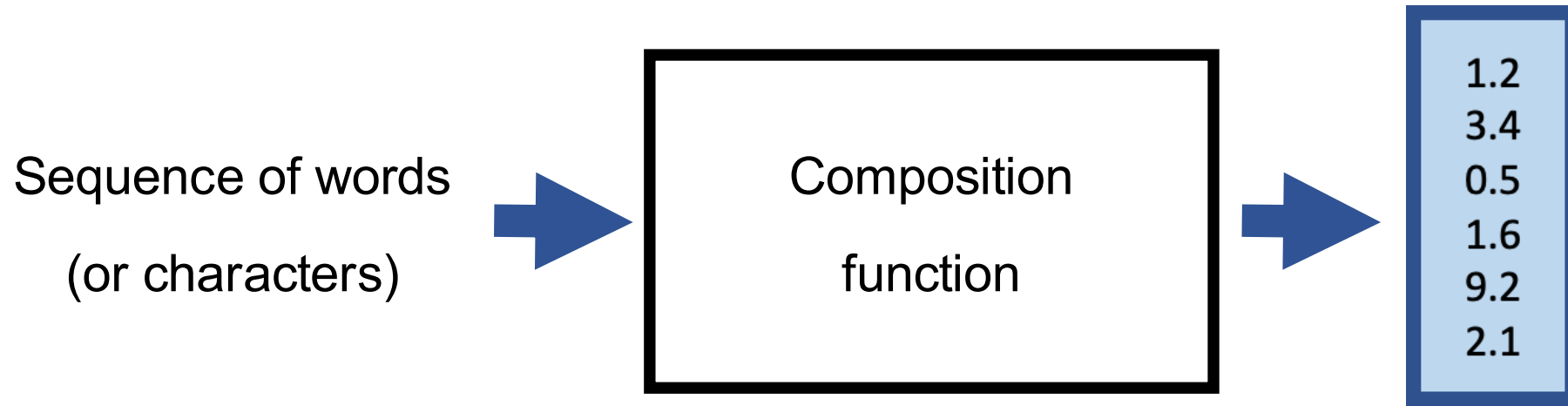


Bag of Words (BOW)
Window BoW (bag of ngrams)
RNN
Transformers
CNN
GraphNN

How to **compose** language?

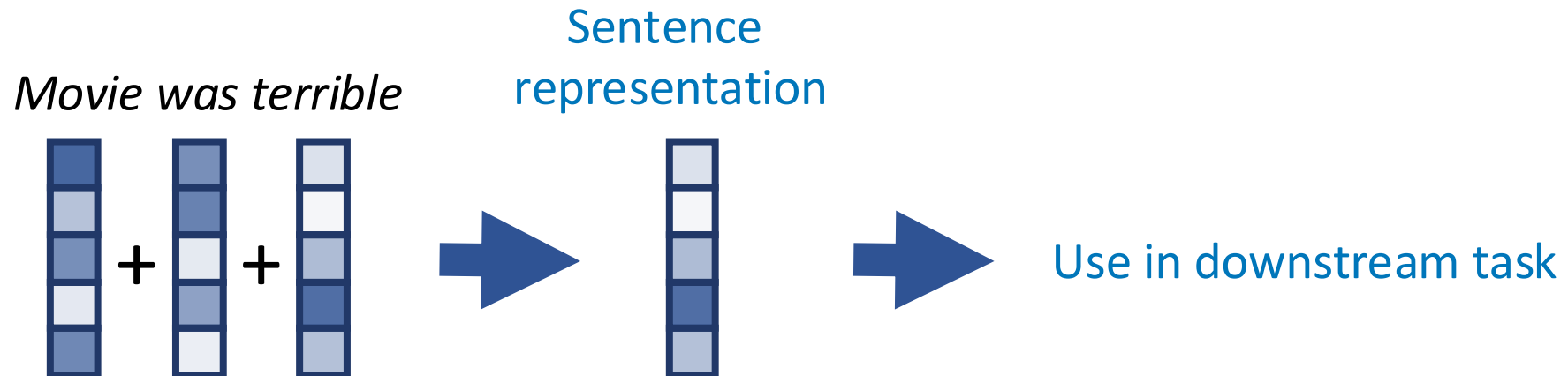


How to **compose** language?



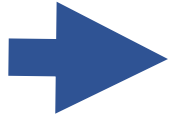
Simplest: take weighted average

Example:



How to use the learned representation?

What do we do with this vector representation after we get it?



Use in downstream task

Vector
representation

Do prediction: classify as spam/not spam

Use to generate text (also known as **decoding**)

Can also be used to generate other stuff (like images!)

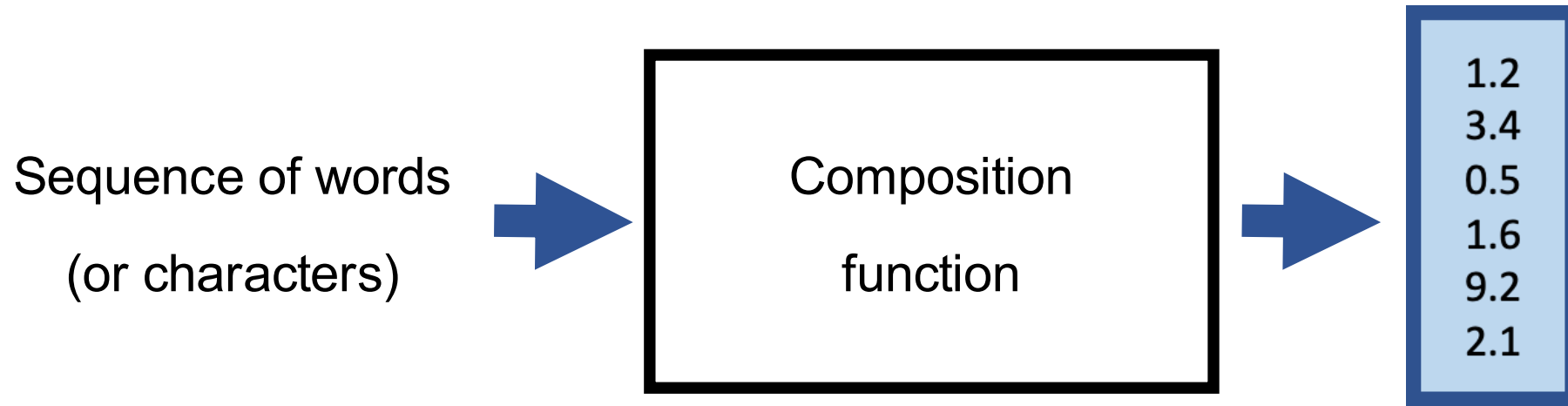
Thank you!



Getty Images

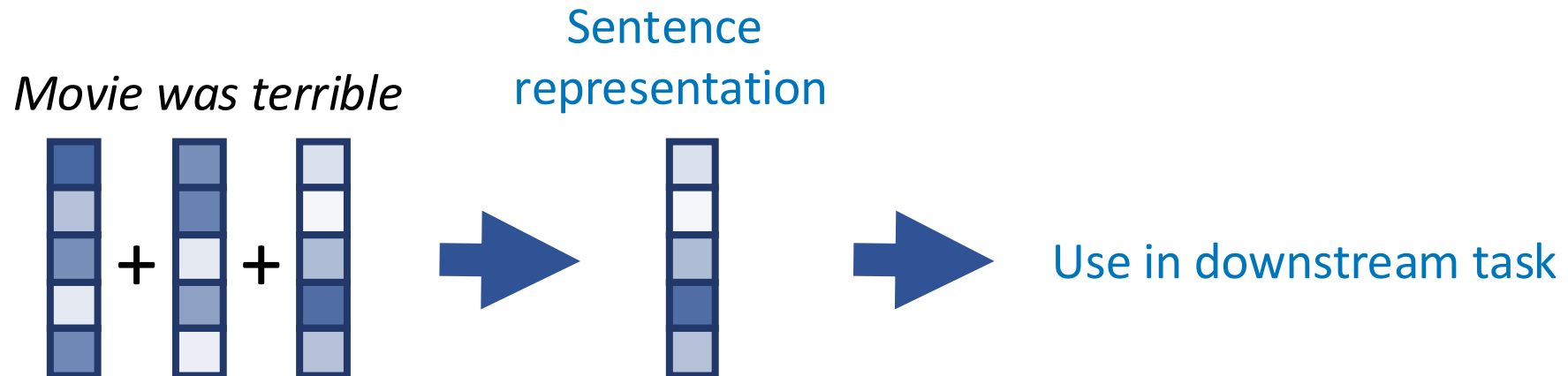
<https://www.thoughtco.com/syntactic-ambiguity-grammar-1692179>

How to **compose** language?



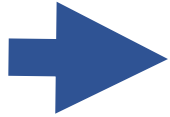
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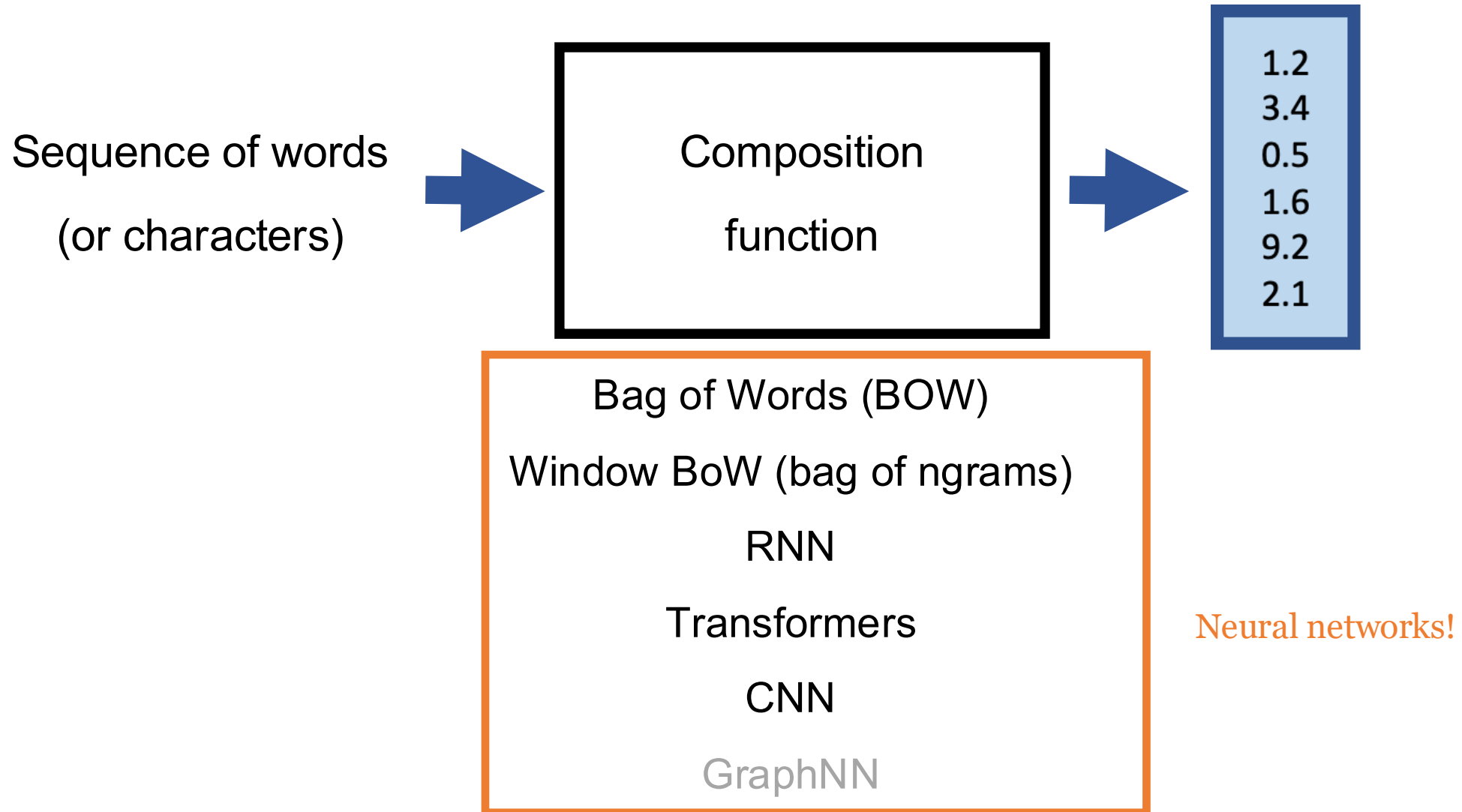
Vector
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How to **compose** language?



Thank you!



Getty Images

<https://www.thoughtco.com/syntactic-ambiguity-grammar-1692179>



ALFRED

A Benchmark for Interpreting
Grounded Instructions for Everyday Tasks

<https://askforalfred.com/>

What does it mean to understand language?
A look at levels of linguistic representations

Translation

Dream of universal translator

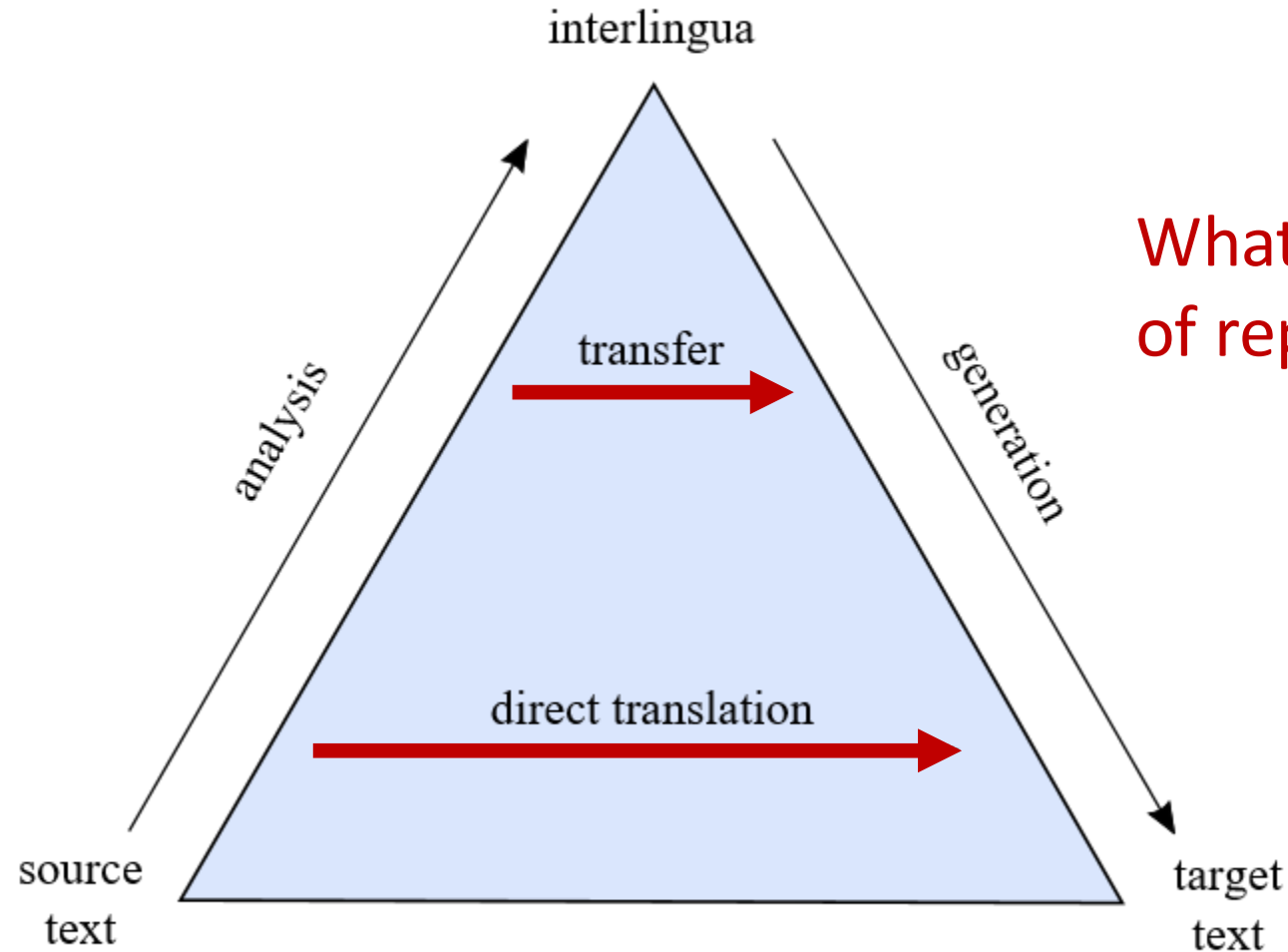
- Different languages: Machine translation

the cat sat on the table  η γάτα κάθισε στο τραπέζι

Not just between languages, but across modalities as well

- Audio to text: Speech recognition
- Image to language: Image captioning

Vauquois Triangle for translation

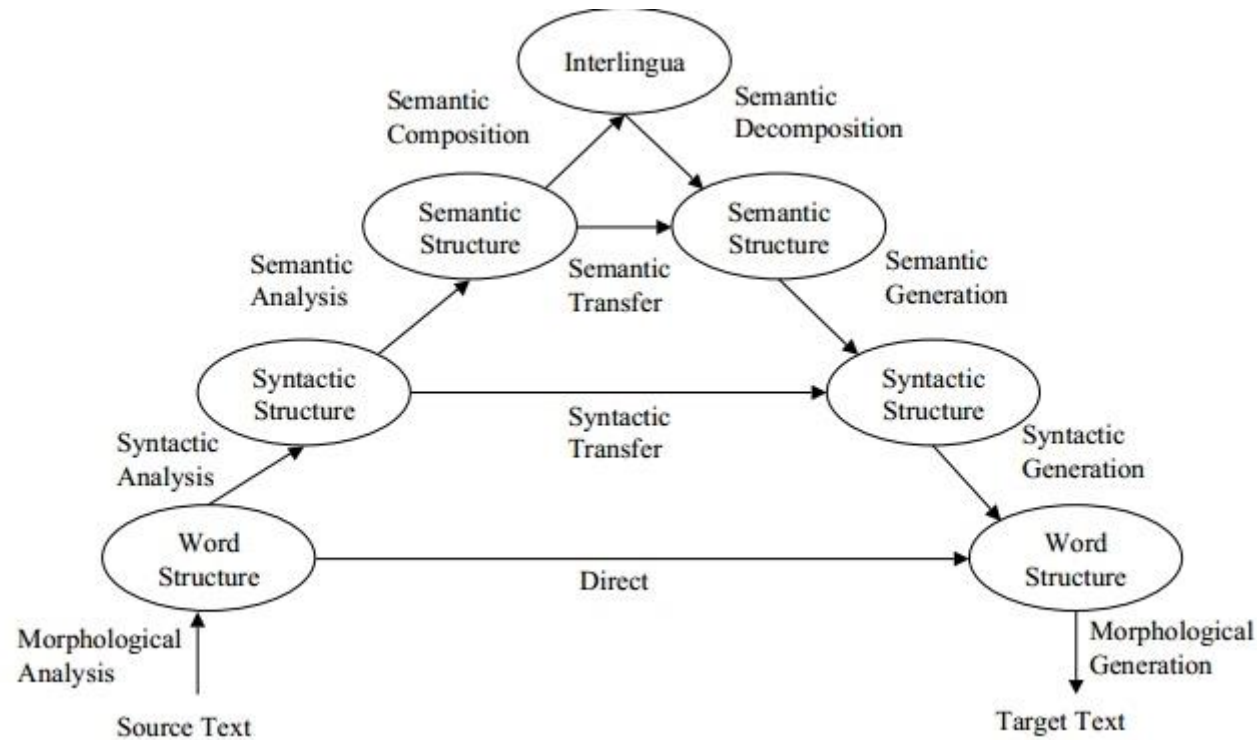


What are the levels of representation?

the cat sat on the table

η γάτα κάθισε στο τραπέζι

Vauquois Triangle for translation



Pragmatics

Semantics

Syntax

Morphology

the cat sat on the table

η γάτα κάθισε στο τραπέζι

Levels of language

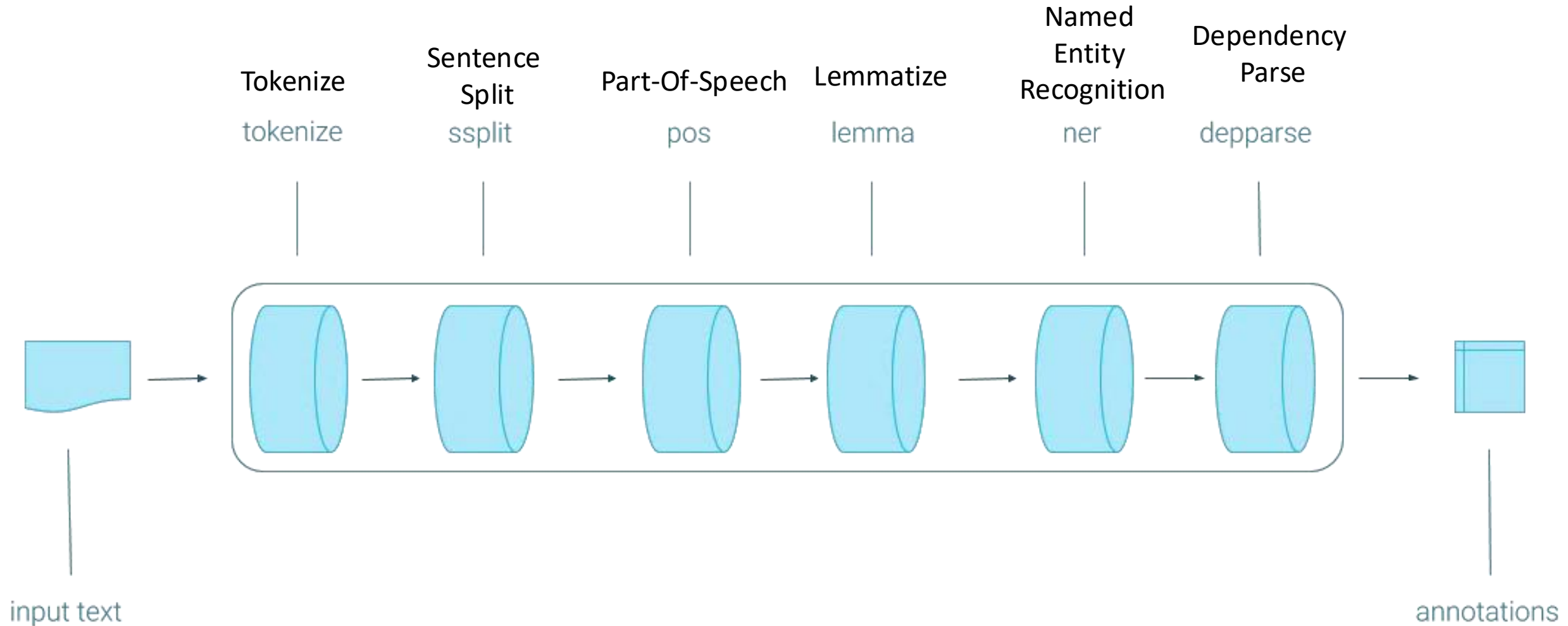
- Phonetics: acoustic and perceptual elements
- Phonology: basic sounds (phonemes) and rules for combination
 - e.g. syllable constraints in Japanese.
 - e.g. English loanword for “dog” in Japanese is “doggu”
- Morphology: how morphemes combine to form words, relationship of phonemes to meaning
 - e.g. delight-ed vs. de-light-ed

Levels of language

- Syntax: set of rules, principles, and processes that govern the structure of sentences in a given language
 - sentence formation
 - o e.g. The clown who the musician hits watches the ballerina
- Semantics: focuses on the relation between *signifiers*, like words, phrases, signs, and symbols, and what they stand for; their denotation
 - meaning (from syntax to logical formulas)
 - e.g. Everyone is not here => what does this mean? Nobody / Not everyone is here.
- Pragmatics: studies the ways in which context contributes to meaning
 - e.g. “The ham sandwich wants another coffee”

More details in optional video lecture (to be posted on Coursys)

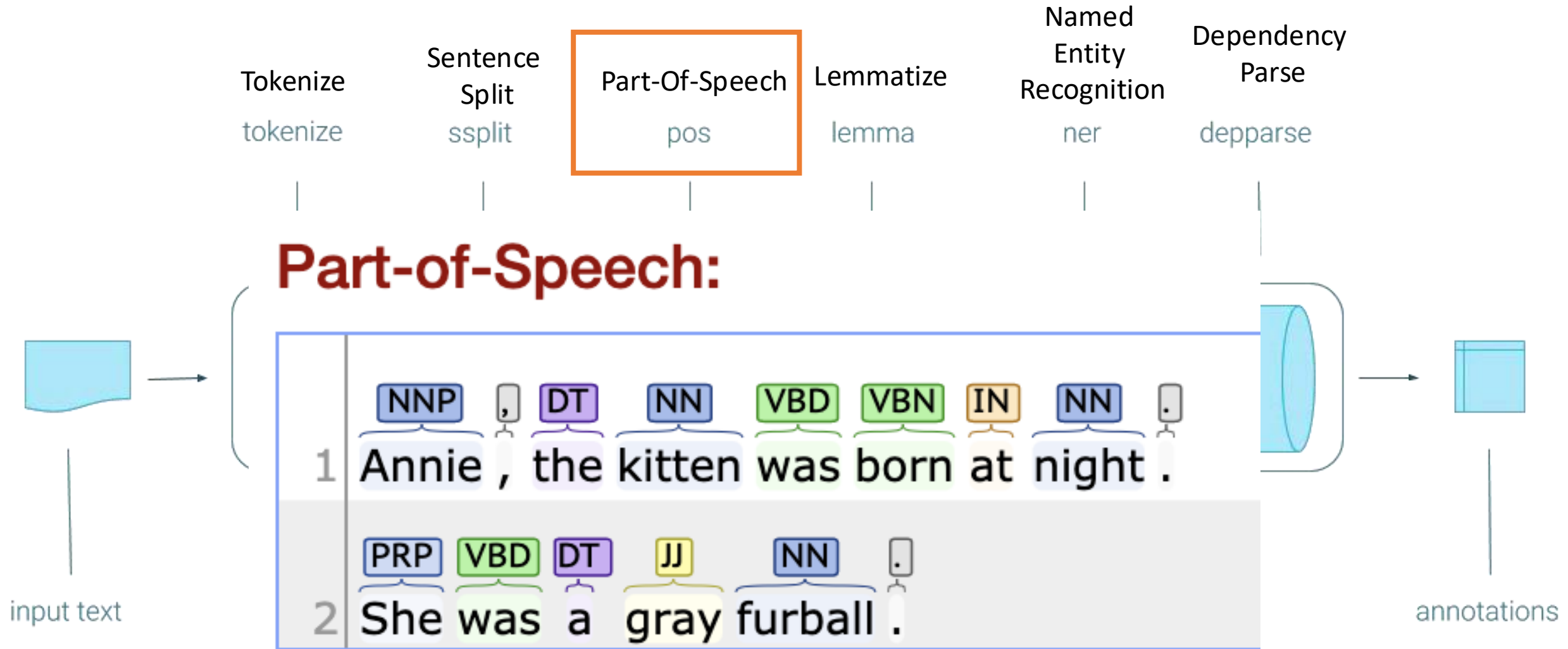
Traditional NLP Pipeline



More details in optional video lecture (to be posted on Coursys)

Annie, the kitten was born at night. She was a gray furball.

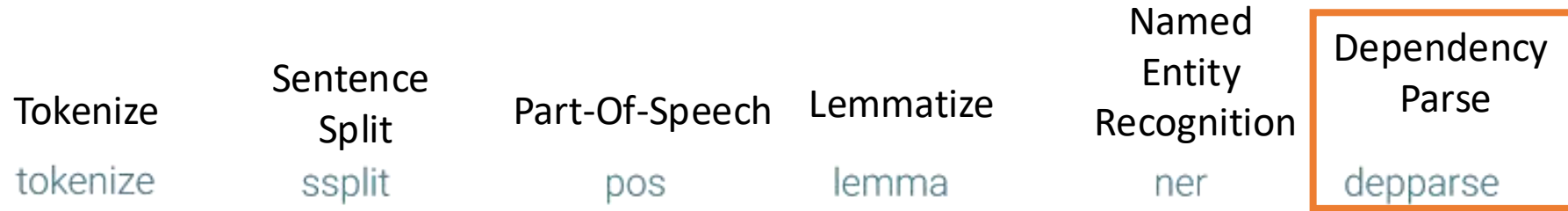
Traditional NLP Pipeline



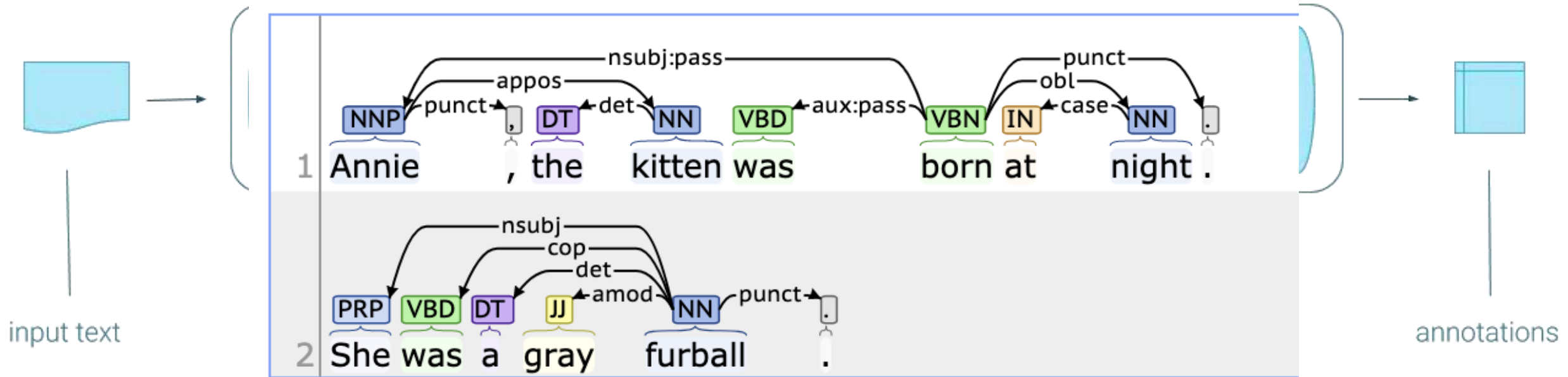
More details in optional video lecture (to be posted on Coursys)

Annie, the kitten was born at night. She was a gray furball.

Traditional NLP Pipeline



Basic Dependencies:



More details in optional video lecture (to be posted on Coursys)

Annie, the kitten was born at night. She was a gray furball.

Try out some NLP systems for yourself!

- Spacy (<https://spacy.io/>)
 - Interoperates with Stanza and AllenNLP
- Stanford CoreNLP / Stanza
 - CoreNLP (<http://stanfordnlp.github.io/CoreNLP/>)
 - Stanza (<https://stanfordnlp.github.io/stanza>)
- AllenNLP (<https://allennlp.org/>)

- Representations
 - Embeddings
 - Compositional representations
 - Structured representations
- Applications

Outline

- **Fundamental Methods and Models**
 - Language Models
 - Classifiers (Naïve Bayes, Logistic Regression, Neural Networks)
 - Text classification
- **Words**
 - Word embeddings
- **Sequences**
 - Sequence modeling
 - Sequence to sequence models and text generation
- **Context**
 - Contextual word embeddings
- **Structures**
 - Syntactic parsing
 - Semantic parsing
- **Applications**
 - Machine Translation
 - Information extraction
 - Question answering
 - Dialogue
 - Multimodal NLP

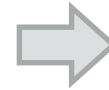
Outline

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 - Dialogue
 - Multimodal NLP

Information Extraction

The Massachusetts Institute of Technology (MIT) is a private research university in Cambridge, Massachusetts, often cited as one of the world's most prestigious universities. Founded in 1861 in response to the increasing industrialization of the United States, ...

Article



*City: Cambridge, MA
Founded: 1861
Mascot: Tim the Beaver
...*

Database

Information Extraction: State of the Art

Dependence on large training sets

ACE: 300K words

Freebase: 24M relations

Not available for many domains (ex. medicine, crime)

Challenging task: even large corpora do not guarantee high performance

~ 75% F1 on relation extraction (ACE)

~ 58% F1 on event extraction (ACE)

Machine comprehension

Amazon_rainforest

The Stanford Question Answering Dataset

The Amazon rainforest (Portuguese: Floresta Amazônica or Amazônia; Spanish: Selva Amazónica, Amazonía or usually Amazonia; French: Forêt amazonienne; Dutch: Amazoneregenwoud), also known in English as Amazonia or the Amazon Jungle, is a moist broadleaf forest that covers most of the Amazon basin of South America. This basin encompasses 7,000,000 square kilometres (2,700,000 sq mi), of which 5,500,000 square kilometres (2,100,000 sq mi) are covered by the rainforest. This region includes territory belonging to nine nations. The majority of the forest is contained within Brazil, with 60% of the rainforest, followed by Peru with 13%, Colombia with 10%, and with minor amounts in Venezuela, Ecuador, Bolivia, Guyana, Suriname and French Guiana. States or departments in four nations contain "Amazonas" in their names. The Amazon represents over half of the planet's remaining rainforests, and comprises the largest and most biodiverse tract of tropical rainforest in the world, with an estimated 390 billion individual trees divided into 16,000 species.

Which name is also used to describe the Amazon rainforest in English?

Ground Truth Answers: also known in English as Amazonia or the Amazon Jungle, Amazonia or the Amazon Jungle Amazonia

Prediction: Amazonia

How many square kilometers of rainforest is covered in the basin?

Ground Truth Answers: 5,500,000 square kilometres (2,100,000 sq mi) are covered by the rainforest. 5,500,000 5,500,000

Prediction: 5,500,000

How many nations control this region in total?

Ground Truth Answers: This region includes territory belonging to nine nations. nine nine

Prediction: nine

How many nations contain "Amazonas" in their names?

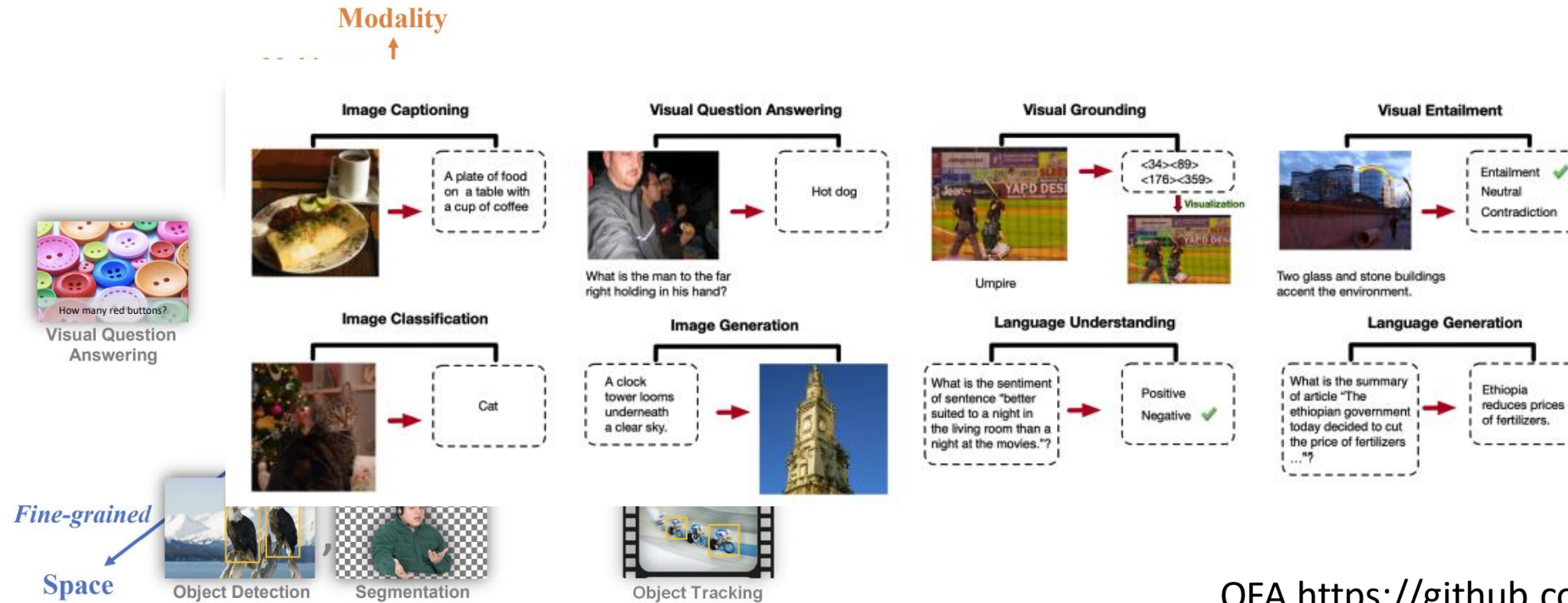
Ground Truth Answers: States or departments in four nations contain "Amazonas" in their names. four four

Prediction: four

What percentage does the Amazon represents in rainforests on the planet?

Multimodal models

Multilingual, answering questions about images, image captioning,



OFA <https://github.com/OFA-Sys/OFA>
(Wang et al, 2022)

Florence

<https://arxiv.org/pdf/2111.11432.pdf>
(Yuan et al, 2021)